

# A STRONGLY APERIODIC MONOTILE IN THREE DIMENSIONS

IOANNIS TSIOKOS

**ABSTRACT.** Socolar and Taylor asked for a single, simply connected three-dimensional prototile that forces nonperiodicity by shape alone, admitting no weakly nonperiodic tiling; the Schmitt–Conway–Danzon biprism and the three-dimensional Socolar–Taylor tile admit screw motions or a periodic stacking direction. We exhibit a rational polyhedral 3-ball  $Q$ , which we call Chair44 (R44): a seven-cube chair whose 24 exposed unit panels carry tiny square-pyramid features, and prove, as a proof submission, that  $Q$  admits tilings of  $\mathbb{R}^3$  by congruent copies, reflections allowed, and that every such tiling has no translational period and a symmetry group of order at most 24; every tiling is homochiral and carries a unique infinite hierarchy of nested supertiles. The solid was designed to a reading of the aperiodic-monotile phenomenon reached with the Six Birds emergence calculus (Section 3.3), and the construction turns on a single finite test, checked by machine: the tile’s own contact rule survives coarsening, so that the decoded parent tiling obeys the tile’s rule and no other. The proof combines a written geometric argument, that the features force every tiling onto a registered lattice, with exhaustive finite enumerations; the companion census is replayed by two independent implementations, every finite gate is kernel-checked in Lean 4 (modulo a named compiler hook per `native_decide` theorem), and the written geometric lemmas and the logical assembly are Lean theorems as well, so that the theorem is kernel-checked modulo the named compiler hooks; the written proofs remain as exposition.

*Proof submission.* The finite checks, the nineteen former hypothesis statements, and the logical assembly of the four tiling clauses are Lean theorems, kernel-checked modulo the named compiler hooks. Section C records the empty hypothesis structure and the dependencies; the written proofs remain as exposition.

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Version 2 DOI: [10.5281/zenodo.22792358](https://doi.org/10.5281/zenodo.22792358)

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*Date:* First published (Version 1): 15 September 2026; Version 2: 16 September 2026 (mathematical baseline commit `d90313a717`).

2020 *Mathematics Subject Classification.* Primary 05B45; Secondary 52C22, 52C23.

*Key words and phrases.* Aperiodic monotile, einstein problem, strongly aperiodic tilings, tilings of three-dimensional space, polyhedral tiles, substitution tilings, hierarchical tilings, limit-periodic model sets, computer-assisted proof, Lean 4 formalization, Six Birds Theory, emergence calculus.

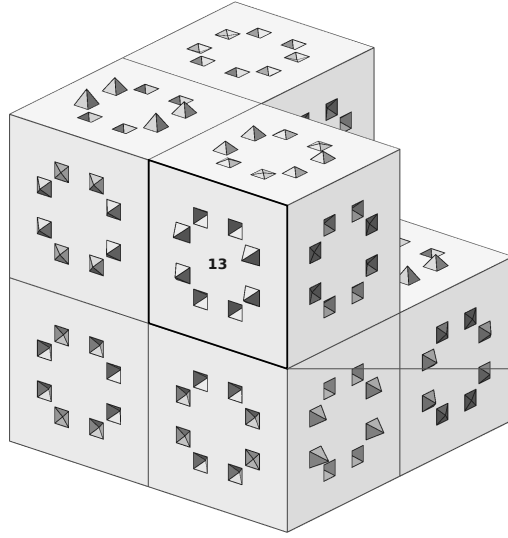
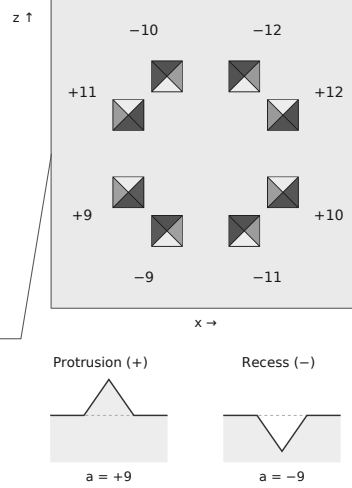
## 1. INTRODUCTION

1.1. **The einstein problem in  $\mathbb{R}^3$ .** Socolar and Taylor, presenting the three-dimensional version of their hexagonal tile, set out what remained to be found: “It is still interesting, however, to search for a single, simply connected 2D or 3D prototile that forces maximal nonperiodicity by shape alone, or one that does not permit any weakly nonperiodic tilings” [40]. Writing after the hat, Kaplan restated the demand from the other side: “Some people regard this screw motion as uncomfortably close to translation, and demand a strongly aperiodic 3D monotile, one that admits tilings whose symmetries never include an infinite cyclic subgroup of any kind. Many of the ideas and algorithms we used to prove the hat’s aperiodicity could be adapted to work in 3D space, but personally I am daunted by the prospect of deducing the behaviour of any candidate shapes that might be discovered there” [21]. The solid  $Q$  of Figure 1 is such a prototile: a closed polyhedral 3-ball with rational vertex coordinates that admits tilings of  $\mathbb{R}^3$  by congruent copies, and every tiling it admits has a finite symmetry group (Theorems 1.1 and 1.2 below).

The study of aperiodic tilings begins with Wang’s tiles and his conjecture that any set of Wang tiles admitting a tiling admits a periodic one [48]; Berger disproved it with an aperiodic set of 20,426 Wang tiles and derived the undecidability of the domino problem [8]. Robinson gave a six-tile aperiodic construction [33] and Penrose a two-tile one [29]; these counts use different permitted motions and matching conventions from Wang tiles, for which Jeandel and Rao proved that the minimum is eleven [20]. The natural endpoint of this reduction is a single shape that admits tilings but only non-periodic ones, an aperiodic monotile or “einstein” [36].

Every candidate before the hat conceded something. Gummelt’s decagon covers the plane only with overlaps [17]; Penrose’s  $1 + \varepsilon + \varepsilon^2$  set lets an arbitrarily high fraction of the area lie in one tile, but the other tiles remain necessary [30]; the Taylor–Socolar hexagon needs markings that let non-adjacent tiles exchange information, and a geometric version of the rule needs a disconnected tile, a tile with cutpoints, or a three-dimensional shape tiling a thickened plane [39–41]; the Walton–Whittaker tile uses orientational rules [47]; and Mampusti and Whittaker say of their dendritic construction that “the monotile we present here is not an einstein in the technical sense” [25]. In a different direction, Greenfeld and Tao disproved the periodic tiling conjecture with a single tile that tiles  $\mathbb{Z}^d$  by translations only, for a dimension  $d$  that “will be extremely large” and with a tile that need not be connected; whether  $\mathbb{Z}^3$  admits such a tile they leave open [15], and Greenfeld and Kolountzakis showed that the tile can be taken connected [14].

In the plane the question is settled. The hat, a polykite, is an aperiodic monotile whose tilings use both the tile and its mirror image [36]; the Spectre tiles aperiodically with a single handedness even when reflections are allowed [37]. Both papers give a hierarchical, computer-assisted proof, and the hat

**Q — seven-cube chair****Panel 13 · eight features**

Exact feature centres and signed height ratios · bases  $\times 5$  · heights  $\times 80$   
 Relief exaggerated for visibility; the carrier retains its true proportions.

FIGURE 1. The solid  $Q$ , with feature bases enlarged  $\times 5$  and signed heights  $\times 80$  for visibility. The seven-cube carrier, feature centres, polarities and relative heights are preserved. The outlined panel 13 and its inset show the same eight features; signed labels give  $a$ , whose true height is  $a/10000$  (positive for a protrusion, negative for a recess). The sections illustrate its  $+9$  and  $-9$  features. The true base side is  $1/50$ ; the drawing is not at true feature scale.

paper a second, computer-free one. Simplified proofs and the structure of the tilings followed quickly [1, 6, 7, 38].

Three dimensions have their own history. The second part of Hilbert's eighteenth problem asked whether anisohedral polyhedra exist in  $\mathbb{R}^3$  [19]; Reinhardt found one [32], Heesch a planar one [18]. The Schmitt–Conway–Danzer biprism, a convex rhombic biprism going back to an unpublished 1988 example of Schmitt [31, 35], admits, with reflections forbidden, only non-periodic tilings: its tilings are periodic layers stacked with a twist, and it admits a tiling whose symmetry group contains a screw motion; with reflections allowed it admits a periodic tiling; Socolar and Taylor call the structure “heterogeneously periodic” [40]. Their own three-dimensional tile enforces the hexagonal matching rules by shape alone and can be made a topological sphere, but its tilings form “corrugated slabs of the limit-periodic pattern that stack periodically in the direction orthogonal to the nonperiodic tiling plane” [39], which makes it “a weakly nonperiodic tiling by Goodman–Strauss’s definition” [40]. Fletcher turned Čulík and Kari’s set of 21 Wang

cubes [46] into a single cubic prototile whose 21 types are 21 orientations, but the matching rules become an atlas of legal one-corona patches rather than the shape of the tile [11]. Kaplan’s survey calls the biprism “both an existence proof and a cautionary tale about the definition of aperiodicity in 3D space” [21].

That definition is the point on which the three-dimensional problem turns. The hat paper, “following Mozes”, calls a set of tiles *strongly aperiodic* if it admits tilings but none with any infinite cyclic symmetry, and *weakly aperiodic* if it admits tilings but none with a cocompact discrete group of symmetries [27, 36]. In the Euclidean plane the two notions coincide for normal tiles [16, Theorem 3.7.1]; in the hyperbolic plane weak aperiodicity comes readily, as with Böröczky’s monotile [9], while Goodman-Strauss constructed a strongly aperiodic set there [12]. In  $\mathbb{R}^3$  the notions differ: with reflections forbidden, the biprism is weakly but not strongly aperiodic. We prove the strong form: every tiling by  $Q$  has a symmetry group of order at most 24, and a finite group has no element of infinite order (Corollary 7.3). Coulbois, Gajardo, Guillon and Lutfalla use a finer taxonomy in which a tile all of whose tilings have finite symmetry group is *mildly* aperiodic and “strongly aperiodic” is reserved for trivial stabilizers; in their terms the hat is mildly aperiodic, “strongly aperiodic monotiles are unknown in all settings”, and their Question 32 asks for one [10]. In their terms  $Q$  is mildly aperiodic and not strongly aperiodic: an explicit tiling by  $Q$  has symmetry group of order eight, and others have trivial symmetry group (Sections 7 and 8.2).

$Q$  is not convex, as the biprism is, and it is not simple: it is a seven-cube chair whose 24 exposed unit panels each carry eight square pyramids of height between  $1/10000$  and  $12/10000$  (Section 2). It is, however, a closed topological 3-ball with rational coordinates, and no matching rule of any kind is imposed: the tilings in Theorem 1.2 are all the ways congruent copies of the solid can fill space without overlapping.

**1.2. Main result.** In this paper we prove the following, as a proof submission in the sense explained at the end of this subsection.

**Theorem 1.1.** *The solid  $Q$  of Figure 1 is a strongly aperiodic monotile of  $\mathbb{R}^3$ :  $Q$  admits tilings of  $\mathbb{R}^3$  by congruent copies, and no such tiling has a symmetry of infinite order.*

**Theorem 1.2.** *Let  $Q$  be the rational polyhedral solid of Section 2. Then*

- (1) *there is a tiling of  $\mathbb{R}^3$  by isometric copies of  $Q$ ;*
- (2) *for every tiling  $T$  by isometric copies of  $Q$ ,  $\text{Per}(T) = \{0\}$  and  $|\text{Sym}(T)| \leq 24$ ;*
- (3) *every tiling  $T$  is homochiral;*
- (4) *in every tiling  $T$  each tile lies in a unique infinite hierarchy of  $2^n$ -scaled chair supertiles in registered poses, nested through the eight child poses.*

*Isometries include reflections. No face-to-face, lattice, common-orientation, connected-contact-graph or local-finiteness hypothesis is imposed.*

Here  $\text{Per}(T)$  is the set of translation vectors  $v$  with  $T + v = T$  and  $\text{Sym}(T)$  the group of isometries of  $\mathbb{R}^3$  that permute the tiles of  $T$  (Section 1.5). The absence of hypotheses in Theorem 1.2 is deliberate. A tiling is any family of isometric copies of  $Q$  with pairwise disjoint interiors whose union is  $\mathbb{R}^3$ : the copies need not meet face-to-face, need not sit on a lattice, need not share an orientation, and their contact graph need not be connected. Local finiteness, which the hat paper notes is automatic for monohedral plane tilings by closed disks, is proved here from the geometry of  $Q$  (Section 4) rather than assumed.

Reflections are allowed in Theorem 1.2, and clause (3) says what they do: nothing. By longstanding tradition in the tiling literature, “shapes are considered congruent if they are equivalent under any Euclidean isometry, including those that reverse orientation” [36], and the hat’s status as a monotile depends on that convention, since every hat tiling uses both handednesses.  $Q$  needs no such caveat. Its tilings are *homochiral* in the Spectre paper’s sense, “for every pair  $T_i, T_j \in \mathcal{T}$ , there is an orientation-preserving isometry mapping  $T_i$  to  $T_j$ ” [37]: a tiling by  $Q$  is wholly left-handed or wholly right-handed, and a reflected and an unreflected copy never occur in the same tiling. By the same paper’s definition, a tile that admits only homochiral non-periodic tilings is a *strictly chiral aperiodic monotile*;  $Q$  is one, in  $\mathbb{R}^3$  (Corollary 7.5). The Schmitt–Conway–Danzer biprism, by contrast, is chiral but admits a periodic tiling once reflections are allowed [40].

Clause (4) is the mechanism. After one ambient isometry every tile of every tiling occupies one of 24 proper cubic frames at an integer position (Section 5); the seven-cube chairs so placed group uniquely into 2-scaled chairs, those into 4-scaled chairs, and so on without end (Section 6). The hierarchy is unique at every level, and that is what forces non-periodicity, as in the Spectre paper [37]. We do not claim that the supertiles of one tile exhaust space: as in the hat paper, tilings in which a tile sits at a corner of its supertile at every level, along “fault lines”, are not excluded (Section 6).

*Proof submission.* The finite parts of the argument, the enumerations of Section 1.4, the written geometric lemmas, and the logical assembly from those lemmas to Theorem 1.2 are theorems checked by the Lean 4 proof assistant, modulo the named compiler hooks of Section C. The written lemmas concern the tube construction, feature companions, global coverage and registration, carrier recovery, and small-collar realization. They are new, written out in full, independently reviewed, and listed by name with the Lean theorems that prove them in Table 3; there are 19 of them. Until the development and this manuscript have been refereed we present the theorem as a proof submission.

**1.3. What an aperiodic monotile is, and how that found  $Q$ .** The proof below follows a reading of the phenomenon that we state first, because it is what produced the solid. Consider a labelling  $x: \mathbb{Z}^3 \rightarrow A$  of the cubic lattice by a finite alphabet. For a nonzero  $p \in \mathbb{Z}^3$ ,  $x$  is  $p$ -periodic exactly

when it factors through  $\mathbb{Z}^3 \rightarrow \mathbb{Z}^3/\langle p \rangle$ ; this quotient is infinite. For tilings by  $Q$  the connection to labellings is registration (Section 5): after one ambient isometry every tile of every tiling sits in one of 24 proper cubic frames at an integer position, and, because  $Q$  has no self-isometry, every translational period of the tiling is then an integer vector (Section 7). We use only this direction; the paper does not identify the periods of a tiling with those of a labelling in any packaged form. For  $Q$ , we prove existence and exclude every nonzero translational period directly, using registration and period halving. The obstruction is productive: the local pieces do fit together globally; what fails is fitting them together periodically.

A hierarchy certifies aperiodicity only when it is forced. Grouping the tiles of a tiling into supertiles proves nothing by itself: on the cubic grid an observer may group cells into  $2 \times 2 \times 2$  blocks in eight ways, and the grouping adds information the tiling does not carry. What proves non-periodicity is a grouping that the tiling determines and that every symmetry of the tiling therefore preserves. Then a symmetry of the tiling is a symmetry of the grouped tiling at every level, so a translation that was a symmetry would survive coarsening to every scale; if each coarsening halves it, it was never there. This is the recognizability, unique-composition and period-halving mechanism used here [13, 37], and we use it as a design target.

With marked tiles the rule that decides which neighbours are legal is external to the pieces; a geometric monotile has to put that rule into what can physically fit. For a marked system with a forced hierarchy to become one bare solid, three things must hold: a solid realizing the marked system exists; every tiling by the solid decodes into a legal configuration of the marked system; and a period of the tiling is a period of the decoded configuration. The second is the one that is easy to lose. It is not enough that the intended tilings survive: every tiling admitted by the bare solid must decode into the intended admissible marked system. One shape does not mean one role: copies of a single solid occupy different positions in the forced structure, and the diversity that a set of marked tiles carries in its catalogue is carried here by the relations between congruent copies.

This reading yields a finite test. Write the legal contacts between neighbouring copies of the solid as a language of contacts, and decode a tiling into the tiling by its parents. The halving argument needs the parent tiling to be again a legal tiling of the same solid, that is, the language of contacts among decoded parents must lie inside the language of contacts among the tiles themselves. Equality of the two languages is the form we check, and it is sufficient for the argument, not necessary: a strictly smaller decoded language would also do. When the decoded language is strictly larger the argument fails, and whether a periodic parent tiling then exists must be checked case by case; for the first bare solid we built, it did (Section 3.3). For  $Q$  the test is one finite identity, verified by machine: the halved parent atlas equals the fine atlas of 44 legal contacts (Section 3.2).

The reading was reached with the Six Birds emergence calculus [43–45]; Section 3.3 quotes its statement of the phenomenon and relates it, clause by clause and at each clause’s level of verification, to the theorems proved here. Nothing in that section is used in the proofs.

The proof is the discharge of that reading, and Figure 2 shows its shape. Existence comes from a substitution: eight rotated copies of the chair form a doubled chair, and iterating the dissection gives a registered tiling of the chair which the features accept (Section 3). Universality comes from the features: each pyramid forces a companion pyramid of the opposite polarity on a neighbouring tile, the companion forces the neighbour’s pose into a discrete set, and an exhaustive census reduces that set to the 44 legal contacts (Section 4); from there every tiling is registered (Section 5), every registered tiling has a unique parent tiling at every level (Section 6), and a translational period would have to halve at every level, hence vanish (Section 7). As in the hat paper, the computer-assisted census is needed only to show that all tilings follow the hierarchy; it is not needed to show that a tiling exists.

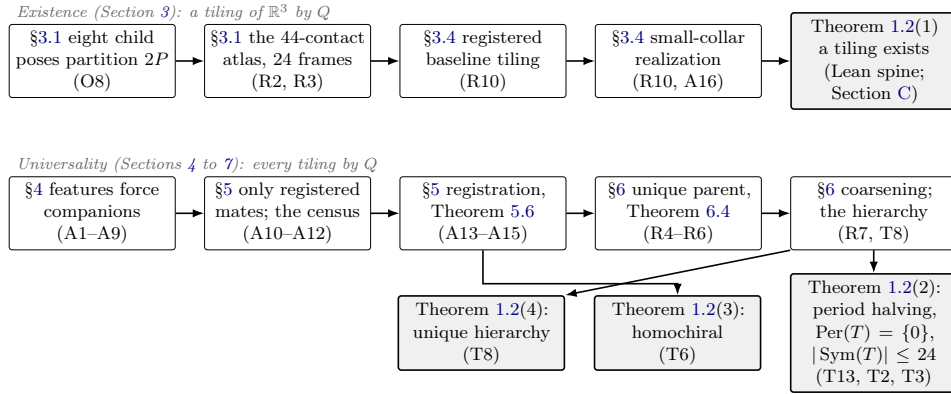


FIGURE 2. Roadmap of the proof. Each box names its section and the rows of the claims ledger (Section C) it rests on.

**1.4. Computer assistance and verification.** Several steps of the proof are exhaustive finite enumerations, listed in Table 6: the 2,388 candidate contact poses of which exactly 44 are legal; the 33 first shells and 28 parent conflicts behind the unique-parent theorem; the  $697 \rightarrow 116 \rightarrow 44$  parent contacts of the coarsening; the companion census of 6,862 mate poses reduced to 44 through 299,975 collision boxes; and the 168-label substitution with its primitivity exponent and modular coincidence depth. The companion census, the largest of these, was checked by two independent Python implementations. Every enumeration is stated as a theorem in Lean 4 and checked by its kernel: those that decide directly by `decide`, and those too large for the kernel’s evaluator by `native_decide`, which adds, per theorem, one named compiler trust hook that we list in Section C. No floating-point number

enters any checked statement: certificates are integers and exact rationals, the Python replay uses exact arithmetic, and Lean decides integer literals. The logical assembly from the enumerations and the named written lemmas to Theorem 1.2 is itself a Lean theorem, and the written lemmas of Sections 3 to 5 are themselves Lean theorems, so that the main theorem is kernel-checked modulo the named compiler hooks listed in Section C; the written proofs remain as exposition. Negative controls (must-fail variants of the inputs and of the formal statements, each checked to fail as expected) accompany the positive build. A one-command replay of the enumerations from the Python standard library runs in about a minute (Section A). The only comparable formalization we know of is Myers’s staging repository for the hat and Spectre, still in progress [28, 42]. *Code.* The repository is given in Section A; its release tag is `paper-v1` (intended; tag creation and licensing remain pending owner actions).

**1.5. Terminology.** Terminology follows Grünbaum and Shephard as the hat paper uses it [16, 36], with “closed topological disk” replaced by “closed topological 3-ball”. A tile is a closed set; a tiling by a set of tiles is a collection of images of tiles under isometries with pairwise disjoint interiors whose union is the whole space; the set, or the single tile, *admits* the tiling. A tiling is monohedral if all its tiles are congruent, congruences including mirror reflections, and locally finite if every ball meets only finitely many tiles. A *monotile* is a tile admitting a monohedral tiling; adapting the hat paper’s reservation of the terms to the present setting, an *aperiodic monotile* or *einstein* here is a closed topological 3-ball admitting tilings but only non-periodic ones, by geometry alone, without any non-geometric matching rule constraining tile adjacencies. *Registered* tilings, the *atlas* of legal contacts, and the *features*, *panels* and *carrier* of  $Q$  are our own terms, defined in Sections 2 and 3; a *supertile of level  $n$*  is a  $2^n$ -scaled chair in a registered pose (Section 6). Here a patch means a finite subfamily of a packing; unlike the hat paper’s disk-patch convention [36], we impose no topological-ball condition. A 1-corona consists of the tiles touching the chosen tile [11]. The notation is collected in Section F, Table 9.

We quote the hat paper’s definitions. “The symmetry group of a tiling is the group of those isometries that act as a permutation on the tiles of the tiling. A tiling is weakly periodic if its symmetry group has an element of infinite order . . . A tiling is strongly periodic if the symmetry group has a discrete subgroup with cocompact action on the space tiled . . . A set of tiles (or a single tile) is weakly aperiodic if it admits a tiling but does not admit a strongly periodic tiling, and strongly aperiodic if it admits a tiling but does not admit a weakly periodic tiling.” And: “Following Mozes [Moz97], we say a set of tiles is strongly aperiodic if it admits tilings but none with any infinite cyclic symmetry” [27, 36]. Since a finite group has no element of infinite order, “strongly aperiodic” in this sense is what Theorem 1.2(2) gives. In the finer taxonomy of Coulbois, Gajardo, Guillon and Lutfalla [10,

§2.1] a tile all of whose tilings have finite symmetry group is *mildly* aperiodic, and *strongly* aperiodic is reserved for trivial stabilizers; in their terms  $Q$  is mildly aperiodic, and it is not strongly aperiodic in that stricter sense, since an explicit tiling has symmetry group of order eight (Section 7). They also distinguish the symmetry group of a tiling from the stabilizer of its family of placements; the two coincide when the tile has no nonidentity self-isometry, as  $Q$  has none (Section E.3), and we write  $\text{Sym}(T)$  throughout.

From the Spectre paper we take, verbatim, that “a monohedral tiling  $\mathcal{T}$  is a homochiral tiling if for every pair  $T_i, T_j \in \mathcal{T}$ , there is an orientation-preserving isometry mapping  $T_i$  to  $T_j$ ” and that a strictly chiral aperiodic monotile is “a tile that admits only homochiral non-periodic tilings”; and that “a set of tiles is hierarchical if, in every tiling admitted by those tiles, every tile is nested within an infinite hierarchy of ever-larger supertiles. If these hierarchies are uniquely determined, then the tilings that contain them must be non-periodic” [37]. Fault lines are defined in Section 6, following the hat paper.

## 2. THE SOLID CHAIR44 (R44), $Q$

**2.1. Start with the chair.** Take a  $2 \times 2 \times 2$  block of unit cubes and remove one corner cube. The remaining seven cubes form the *carrier*,  $P$ . Chair44 is this chair with eight small bumps or dents on each exposed unit square. We call those squares *panels*; there are 24 in all, including the three inside the notch. Figure 3 gives the complete panel recipe. Its small signed numbers tell you which feature to put at each position:  $+j$  means a pyramid pointing out of the solid, and  $-j$  means a pyramid-shaped dent of the same depth. A bump and a matching dent have equal magnitudes and opposite signs.

Fix coordinates by taking the removed cube to be  $[1, 2]^3$ . Thus

$$P = \bigcup_{a \in \{0,1\}^3, a \neq (1,1,1)} (a + [0, 1]^3).$$

The six outer faces contribute  $4 + 4 + 4 + 3 + 3 + 3 = 21$  panels; the notch contributes three more. The large numbers  $0, \dots, 23$  in the diagram identify panels, not heights. They retain the numbering of the exact solid file.

**2.2. Read the panel diagram.** Each of the six diagrams in Figure 3 collects panels with one outward normal. The two axes named above a diagram increase to the right and upwards, respectively. For example, “normal  $-y$ ; axes  $x, z$ ” means that  $x$  increases rightwards and  $z$  upwards. The layouts use these global coordinates. To compare them with a physical model, look at a panel from outside with the second named axis pointing up. The diagrams for normals  $+x, -y$  and  $+z$  then look as drawn; those for  $-x, +y$  and  $-z$  look reflected left to right. This change of viewpoint does not change which feature belongs at each coordinate.

To locate a panel on the chair, read its position in that diagram’s  $2 \times 2$  grid. A left or bottom square has centre coordinate  $1/2$  along the corresponding axis;

a right or top square has centre coordinate  $3/2$ . The remaining coordinate is 0 for a negative outward normal and 2 for a positive one, except for the three notch panels: panel 4 lies at  $x = 1$ , panel 14 at  $y = 1$ , and panel 23 at  $z = 1$ . This locates every panel and fixes the in-panel positions of all eight features. In particular, the square in the upper right of each positive-normal diagram is an inset notch panel, not a face at the missing outer corner.

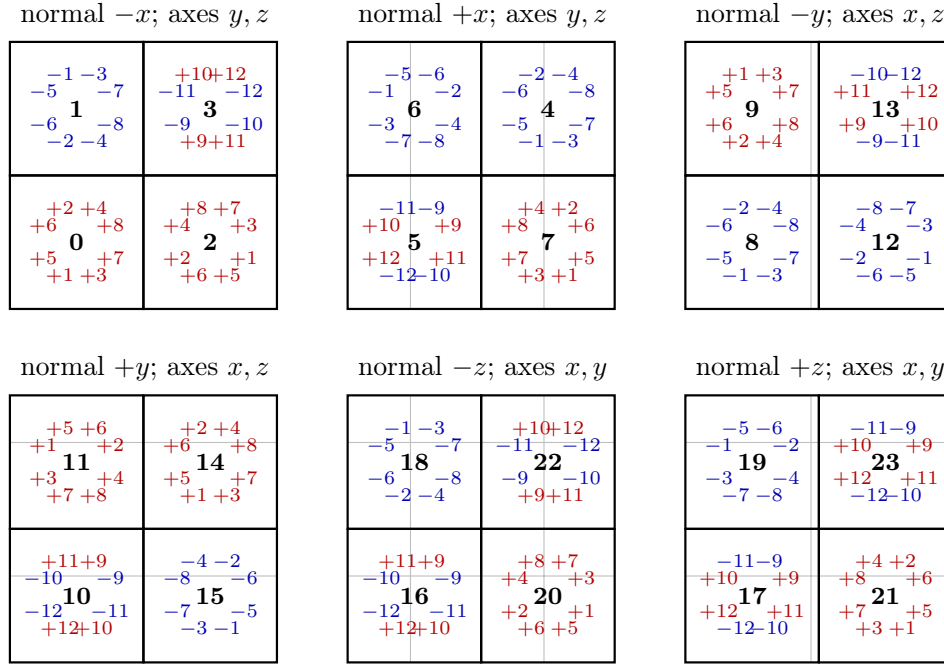


FIGURE 3. Complete panel recipe for Chair44, with the original panel IDs and signed heights. In each grid the first named axis increases rightwards and the second upwards. The large central number is the panel ID; the eight small signed numbers are feature coefficients. Panels 4, 14, 23 are the three inset faces of the notch. Positive coefficients denote bumps and negative coefficients denote dents. The panels are laid out in global coordinates; the diagrams are not exterior views.

You can make a marked paper model by copying these signed numbers onto the corresponding squares of a seven-cube chair, keeping the indicated coordinate orientation. That model is a guide to the shape. The actual tile has no printed labels or imposed matching rules: the pyramids themselves determine what can fit. When two panels meet face to face, their outward normals point in opposite directions. Compare the features at the points of space where they coincide: at each such point a bump must meet a dent of the same magnitude.

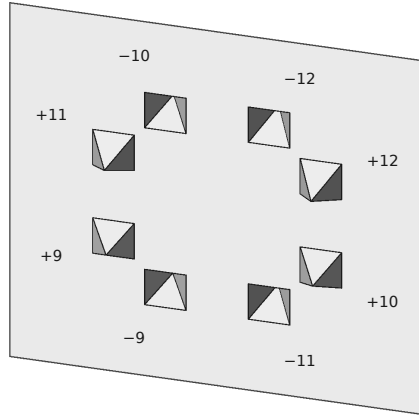
**2.3. From the labels to the solid.** All panels use the same eight feature positions. Relative to the panel centre, in the order of the two axes named in the diagram, these are

$$\left(\pm\frac{1}{8}, \pm\frac{1}{4}\right), \quad \left(\pm\frac{1}{4}, \pm\frac{1}{8}\right).$$

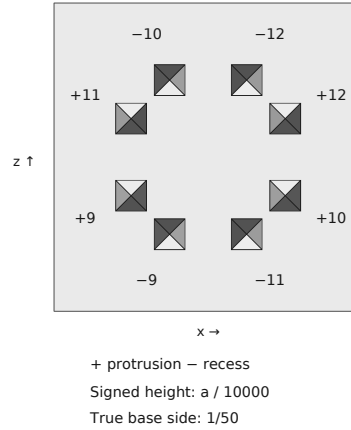
Each feature has an axis-parallel square base of side  $1/50$  (half-width  $\eta = 1/100$ ). Its signed height along the outward normal is  $a/10000$ , where  $a$  is the small signed number at that position. The rest of the panel stays flat. These instructions specify all 192 features without requiring a mesh file.

*For example, read panel 13.* It is the upper-right square of the  $-y$  diagram, so its centre is  $(3/2, 0, 3/2)$  and its in-panel axes are  $x, z$ . The coefficient at offset  $(-1/8, -1/4)$  is  $-9$ . Its base centre is therefore  $(11/8, 0, 5/4)$ , and its apex is  $(11/8, 9/10000, 5/4)$ : since the outward normal is  $-y$ , the negative sign cuts a dent into the solid, towards positive  $y$ . The same panel's  $+9$  at offset  $(-1/4, -1/8)$  points outwards instead. Figure 4 shows both the plan and oblique views of this panel.

**One exposed panel · panel 13**



**Exact arrangement**



Same eight features as the highlighted panel in Figure 1.  
Centres unchanged · bases  $\times 5$  · heights  $\times 80$  · signed height ratios preserved

FIGURE 4. Panel 13, the panel outlined in Figure 1, with its eight square-pyramid features. The outward normal is  $-y$  and the plan view uses global axes  $x, z$ . Labels are the same signed coefficients as in Figure 3. For visibility, bases are enlarged  $\times 5$  and signed heights  $\times 80$  about their unchanged centres; the true base side is  $1/50$  and the true signed height is  $a/10000$ .

**2.4. Why these features?** The bare chair can tile periodically. The features distinguish arrangements that its flat panels alone cannot distinguish. Their heights were chosen so that the contacts needed by the eight-child construction of Section 3.1 fit, including the contacts created when that construction is repeated. Each required pair of coincident features imposes an equation  $a_u = -a_v$ . The resulting equations leave twelve independent signed groups; assigning them the distinct magnitudes  $1, \dots, 12$  gives the height table in Figure 3. The exact reconstruction of that table is in Section E.1. Proving that *every* tiling follows the intended hierarchy requires the registration and recognition arguments in Sections 5 and 6; the panel recipe alone does not prove it.

The numerical dimensions make the features small and separate while retaining the distinct angles used to force registration. Their purpose in the proof is explained in Sections E.4 and E.5; readers following the construction can keep the single height rule  $a/10000$  in mind. The resulting solid has rational vertices, volume 7 and the topology of a closed 3-ball, and no nonidentity self-isometry. Its volume is unchanged because the bump and dent volumes cancel. The topology, asymmetry and mesh checks are recorded in Section E. The parameters can also vary within the stated open conditions (Section E.6); the explicit solid described here uses the fixed values above.

### 3. FINDING $Q$

**3.1. The substitution.** The chair is a rep-tile: eight copies of  $P$  form the doubled chair  $2P$ . Write a *frame* as a pair  $(p, s)$  of a permutation of the coordinate axes and a sign vector, acting by  $G(x)_i = s_i x_{p_i}$ ; a *pose* is a frame followed by a translation. The eight child poses are listed in Table 1 and drawn in Figure 5. Every child frame is a proper rotation, and the 56 unit cubes of the eight children partition  $2P$  exactly, as sets of integer cells (a Lean theorem by kernel `decide`). Refinement acts on poses by

$$(G, t) \mapsto \{(GH, 2t + Gu) : (H, u) \text{ a child pose}\},$$

Thus refinement replaces the doubled support of a chair by eight unit-size chairs; iteration from the native chair produces patches of 8, 64, 512, ... chairs (Figure 6).

Call a pose *registered* if its frame is one of the 24 proper cubic rotations and its translation is an integer vector, and call a tiling *registered* if, after one ambient isometry, every tile is in a registered pose. In a registered tiling a chair with its pose normalized to the identity has 22 grid cells adjacent to it across its 24 panels, and a neighbouring chair covering one of them can sit in any of the 48 signed frames; subtracting its seven cells from each shell cell and rejecting overlaps of the two chairs leaves 2,388 candidate face-neighbour poses. The features decide which of them are legal: two pyramids fit only as a protrusion into a recess of the same magnitude, and because every one of the twelve magnitudes is used by exactly one signed component, equality of the two profiles across a shared panel is checked exactly, centre by centre. Exactly

TABLE 1. The eight child poses of the doubled chair (permutation  $p$ , signs  $s$ , translation  $u$ , in the coordinates of  $2P$ ). Every frame is a proper rotation.

| child   | $p$       | $s$       | $u$       |
|---------|-----------|-----------|-----------|
| 000     | (0, 1, 2) | (+, +, +) | (0, 0, 0) |
| 001     | (1, 0, 2) | (+, +, -) | (0, 0, 4) |
| 010     | (0, 2, 1) | (+, -, +) | (0, 4, 0) |
| 011     | (2, 0, 1) | (+, -, -) | (0, 4, 4) |
| 100     | (2, 1, 0) | (-, +, +) | (4, 0, 0) |
| 101     | (1, 2, 0) | (-, +, -) | (4, 0, 4) |
| 110     | (0, 1, 2) | (-, -, +) | (4, 4, 0) |
| central | (0, 1, 2) | (+, +, +) | (1, 1, 1) |

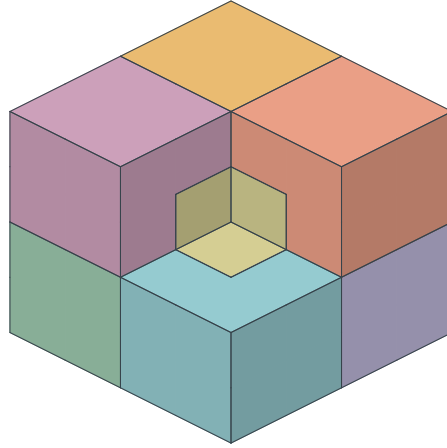


FIGURE 5. The eight-child dissection: eight properly rotated chair carriers partition  $2P$ . Colours distinguish children; the small boundary features are omitted.

44 of the 2,388 poses are legal (a Lean theorem using `native_decide`). We call this set the 44-*contact atlas*  $\mathcal{A}_{44}$  (Figure 7). Its 44 poses use 19 distinct relative rotations, and those 19 generate the whole 24-element proper cubic group (`native_decide`). The word “atlas” is Fletcher’s, who uses it for a matching rule imposed on a cubic prototile [11]; here it names a theorem about what the shape permits, and no rule is imposed. To see the substitution and the atlas in action, the repository includes an interactive browser-based viewer of the solid and its patches.

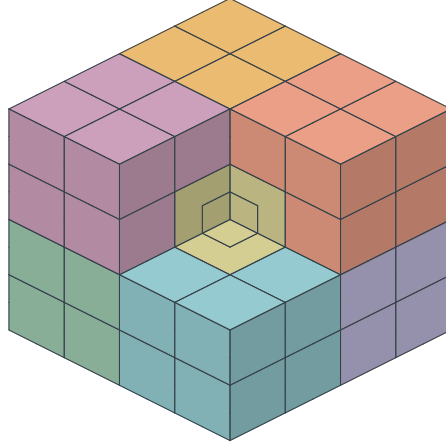


FIGURE 6. Two refinements: 64 chair carriers partition  $4P$ , coloured by their eight parents in the same view as Figure 5. Thin lines separate children; thicker lines mark parent boundaries and carrier creases. Boundary features are omitted; colour is annotation only.

**3.2. The design test in finite form.** The test of Section 1.3 asks whether the decoded parents obey the tile’s own rule. In a registered tiling whose chairs have been grouped into doubled chairs, a legal contact between two fine chairs together with a choice of which child each is in its parent determines the relative pose of the two parents. Enumerating the  $8 \times 44 \times 8$  possibilities and removing duplicates gives 697 candidate parent contacts; 116 of them have disjoint parent bodies; and testing every fine contact across the two parents leaves 44 legal parent contacts, every one with an even translation. Dividing the translations by two gives *exactly* the fine atlas  $\mathcal{A}_{44}$ , not merely a set of the same size:

$$\frac{1}{2} (\text{admitted parent atlas}) = \mathcal{A}_{44}$$

(a Lean theorem using `native_decide`). This is the identity the whole construction was designed to satisfy. Together with unique parent recognition and small-collar realization, this identity permits coarsening (Section 6): divide parent positions and lengths by two and replace each parent carrier by a fresh copy of  $Q$ . Translation covariance sends a period  $p$  to  $p/2$ , so Section 7’s halving argument applies at every level. The fact that the 19 contact rotations generate the full proper cubic group, rather than a subgroup, is what makes the parent frames the same set of frames as the child frames.

| #  | frame $(p) (s)$     | offset       | #  | frame $(p) (s)$     | offset      |
|----|---------------------|--------------|----|---------------------|-------------|
| 1  | (0, 1, 2) (-, -, +) | (2, 2, -2)   | 23 | (1, 0, 2) (+, +, -) | (1, 1, 3)   |
| 2  | (0, 1, 2) (-, -, +) | (2, 2, 2)    | 24 | (1, 2, 0) (-, -, +) | (2, 2, -2)  |
| 3  | (0, 1, 2) (-, -, +) | (3, 3, -1)   | 25 | (1, 2, 0) (-, -, +) | (2, 2, 2)   |
| 4  | (0, 1, 2) (-, -, +) | (3, 3, 1)    | 26 | (1, 2, 0) (-, -, +) | (3, 3, 1)   |
| 5  | (0, 1, 2) (-, +, -) | (2, -2, 2)   | 27 | (1, 2, 0) (-, +, -) | (2, -2, 2)  |
| 6  | (0, 1, 2) (-, +, -) | (2, 2, 2)    | 28 | (1, 2, 0) (-, +, -) | (2, 2, 2)   |
| 7  | (0, 1, 2) (+, -, -) | (-2, 2, 2)   | 29 | (1, 2, 0) (-, +, -) | (3, -1, 3)  |
| 8  | (0, 1, 2) (+, -, -) | (2, 2, 2)    | 30 | (1, 2, 0) (+, -, -) | (-2, 2, 2)  |
| 9  | (0, 1, 2) (+, +, +) | (-1, -1, -1) | 31 | (1, 2, 0) (+, -, -) | (2, 2, 2)   |
| 10 | (0, 1, 2) (+, +, +) | (1, 1, 1)    | 32 | (2, 0, 1) (-, -, +) | (2, 2, -2)  |
| 11 | (0, 2, 1) (-, +, +) | (4, 0, 0)    | 33 | (2, 0, 1) (-, -, +) | (2, 2, 2)   |
| 12 | (0, 2, 1) (+, -, +) | (-1, 3, -1)  | 34 | (2, 0, 1) (-, -, +) | (3, 3, 1)   |
| 13 | (0, 2, 1) (+, -, +) | (0, 4, 0)    | 35 | (2, 0, 1) (-, +, -) | (2, -2, 2)  |
| 14 | (0, 2, 1) (+, +, -) | (0, 0, 4)    | 36 | (2, 0, 1) (-, +, -) | (2, 2, 2)   |
| 15 | (0, 2, 1) (+, +, -) | (1, 1, 3)    | 37 | (2, 0, 1) (+, -, -) | (-2, 2, 2)  |
| 16 | (1, 0, 2) (-, +, +) | (0, 0, 0)    | 38 | (2, 0, 1) (+, -, -) | (-1, 3, 3)  |
| 17 | (1, 0, 2) (-, +, +) | (4, 0, 0)    | 39 | (2, 0, 1) (+, -, -) | (2, 2, 2)   |
| 18 | (1, 0, 2) (+, -, +) | (0, 0, 0)    | 40 | (2, 1, 0) (-, +, +) | (3, -1, -1) |
| 19 | (1, 0, 2) (+, -, +) | (0, 4, 0)    | 41 | (2, 1, 0) (-, +, +) | (4, 0, 0)   |
| 20 | (1, 0, 2) (+, +, -) | (-1, -1, 3)  | 42 | (2, 1, 0) (+, -, +) | (0, 4, 0)   |
| 21 | (1, 0, 2) (+, +, -) | (0, 0, 0)    | 43 | (2, 1, 0) (+, +, -) | (0, 0, 4)   |
| 22 | (1, 0, 2) (+, +, -) | (0, 0, 4)    | 44 | (2, 1, 0) (+, +, -) | (1, 1, 3)   |

FIGURE 7. The 44-contact atlas  $\mathcal{A}_{44}$ .

**3.3. How the tile was found.** This section records how the solid was found; nothing in it is used in the proofs. Three computed steps led to  $Q$ , each recorded in a public artifact of the repository.

*Step 1: a marked chair.* The construction record of the repository (steps 140–165) introduces the seven-cube chair as a rep-8 rep-tile and proves that a single-marking local rule on the chairs, read off a radius-one box, admits the chair substitution; a SAT solver finds no periodic tiling satisfying the rule up to period  $L = 7$ . The record calls the all-scale recursion an “honest gap”: this is a marked candidate with strong finite evidence, not a proved marked einstein. Attempts to realize the marks as notches on the chair failed; the verdict at the time was that the rule is too complex for fixed covariant notches.

*Step 2: a bare solid that is a periodic control.* The first bare solid put 192 square pyramids with signed heights on the 24 panels, in fixed child frames. It admits an all-scale aperiodic hierarchy, and it also admits a legal eight-copy periodic tiling. Decoding its tilings into parent tilings explained why: the parents obey a weaker rule than the tiles, 62 legal fine contacts decoding to 398 parent contacts, so the decoded language is the language of all bare-chair tilings. The packet further proves that a deterministic translation-covariant recoding with uniformly bounded neighbourhood and displacement, if it realizes the specified hierarchical tiling by one compact solid, also realizes a

periodic tiling. Requirement two of Section 1.3 fails, not requirement three: period preservation holds exactly,  $\text{Per}(T) = 2 \text{Per}(D(T))$  for the decoding  $D$ ; what fails is preservation of the solid’s admissibility rule, since the decoded parents obey the less restrictive bare-chair language.

*Step 3: the frame change.* Replacing the fixed child frames by the proper-rotation frames of Table 1, which generate the full 24-element cubic group, and recomputing the profile from the closed contact set ( $21 \rightarrow 30$  contacts) produced  $Q$ , with 44 legal contacts and a parent atlas that halves to the same 44. The decisive finite distinction for the halving test is equality of the decoded parent atlas with the fine atlas,  $44 = 44$  against  $62 \rightarrow 398$ . The unrestricted alignment proof of Sections 4 and 5, that every tiling, not only registered ones, is forced onto the lattice, was written last. These stages are recorded in packets P3–P6 of the repository’s six-packet hash chain, each packet naming its predecessor by hash and replayed and re-verified here; P6 strengthens the unrestricted alignment argument.

*The reading behind the steps.* What follows is the author’s account of the route, in the sense in which the hat paper recounts its own discovery, and it carries no claim. In the author’s recollection, the tile was found by asking what an aperiodic monotile is through the Six Birds framework, an emergence framework developed by the author [43–45], and then having an AI reasoning model that was given the framework design a solid to satisfy that description (see the disclosure after the acknowledgements); the packets that produced  $Q$  name the framework as their structural compass and state their diagnoses of the control in its terms. The framework’s statement of the phenomenon is its law G11, “Local-Rule Global-Anti-Symmetry” [44], stated for a finite local constraint system  $C$  over configurations  $\mathbb{Z}^d \rightarrow A$ :

Assume: [H-G11-hierarchy] every globally admissible  $C$ -configuration carries the declared hierarchy uniquely at every scale; the hierarchy is locally forced by the rules of  $C$ ; and for every nonzero period vector  $p$  some scale  $k$  has forced marker or block structure that cannot be invariant under translation by  $p$ ; [H-G11-nonempty] at least one global configuration satisfies all local rules of  $C$ . Then admissible configurations exist; for every nonzero  $p$  the hierarchy supplies a scale whose forced structure is incompatible with  $p$ -periodicity; because the forcing is local, the incompatibility is witnessed on a finite region  $R_p$ , yielding a per-period defect certificate; and therefore every admissible configuration is aperiodic.

The source lists Berger, Robinson, Penrose, the hat and the Spectre as its instances and reads its own content as “P2 constraint gating, P4 hierarchy across forced scales, and P1 descent obstruction per candidate period”. For  $Q$  the instance is the registered system  $C_{44}$ : labellings of  $\mathbb{Z}^3$  by the 168 registered cell labels (a chair in one of 24 frames at one of seven cells) obeying the atlas  $\mathcal{A}_{44}$ . Registration carries tilings of  $Q$  to  $C_{44}$  (Section 5, resting on written lemmas) and the small-collar realization carries registered baseline tilings back (Section 3.4, a written lemma); we do not claim a packaged

theorem that the two are one object. Clause by clause: [H-G11-nonempty] is Theorem 1.2(1); uniqueness at every scale is Theorem 1.2(4); local forcing is the unique-parent theorem of Section 6 applied level by level, the same finite rule determining the parent from a neighbourhood whose radius grows with the level, which is not local derivability from the unmarked chair tiling; and the period clause is reached not by exhibiting G11's per-period certificates  $R_p$ , which our proof does not extract, but by the halving tower of Section 7. At their verification levels, all four clauses are Lean theorems of the pinned development, kernel-checked modulo the named compiler hooks: existence (`existence`); the hierarchy, with uniqueness among arbitrary geometrically nested registered families (`geometric_hierarchy_canonical`, `geometric_hierarchy_unique`); local forcing, from the finite shell and parent-atlas checks via `native_decide` together with the proved local-to-global assembly and the proved registration bridge; and period exclusion, by the Lean halving tower. The design test of Section 3.2 is, in the framework's terms, its sufficiency and composability gate (laws F7 and F28 of [43]). That reading of the laws is the packets' own; F7 concerns factorization of declared readouts, whereas preservation of legal tilings under coarsening is the separate geometric theorem of Section 6. The catalog contains no tiling instance. Section 9 develops this distinction as a complementary interpretation of the proved construction.

The framework supplied a reading and a test. It proves nothing about  $Q$ : existence is the substitution's, universality is the companion argument's and the census's, and the framework's own wording discipline says that its vocabulary certifies no claim. The monotile theorem is certified by the proof and by the verification record of Section 1.4, and G11 is not a step of any proof here.

### 3.4. Existence.

**Theorem 3.1** (existence; R§9, `existence`, T1N).  *$Q$  admits a tiling of  $\mathbb{R}^3$ .*

*Proof.* Every contact that occurs inside a refined patch lies in the closed set of 30 contacts from which the profile was built (Section E.1), so every finite substitution patch is a packing of copies of  $Q$  whose internal panels match. Two refinements of the native chair contain the same-frame grandchild at  $(2, 2, 2)$ : the central child  $(I, (1, 1, 1))$  has its own 000 child there. Put  $c_n = 2(4^n - 1)/3$  and  $A_n = \sigma^{2n}(P) - c_n(1, 1, 1)$ , the  $2n$ -fold refinement recentred. The grandchild relation makes  $A_n$  a literal subpatch of  $A_{n+1}$ , with the same frames, and the support of  $A_n$  contains the cube  $[-c_n, (4^n + 2)/3]^3$ . These cubes exhaust  $\mathbb{R}^3$ , so the nested union is a registered baseline tiling of the whole space by chairs whose contacts all lie in the closed contact set. The *small-collar realization* lemma then applies: a locally finite baseline chair tiling whose profiles match on every shared panel is carried to a tiling by  $Q$  by the glued tube homeomorphism of Section E.5, each chair going exactly to its copy of  $Q$  because every feature tube belongs to the two cells on either side of its panel, both with unique owners (Section 5, Lemma 5.5).

Small-collar realization is the Lean theorem `small_collar_realization_holds` (T1N); its formal proof identifies duplicate tube sites by their centres, so that the conjugated tube maps agree where two chairs name the same tube, and separates distinct sites by  $21/200 > 1/16$ .  $\square$

The same-frame grandchild, the exact dissection and the first two nested patches (64 and 4,096 tiles; 448 and 28,672 cells) are checked by Lean (`native_decide`); the all- $n$  statement is the induction above, assembled in Lean from those checks and the realization theorem. The construction is explicit; existence is not assumed to make Theorem 1.2(2) non-vacuous, and, as in the hat paper, it is proved independently of the census that establishes universality.

#### 4. FEATURES FORCE COMPANIONS

This section and the next prove that every tiling by  $Q$  is registered (Figure 8). The argument is continuous geometry: the pyramids are tiny, but their edges carry dihedral angles that only a pyramid of the same magnitude and opposite polarity can complete, and that forces every feature of every tile to be matched, whole, by one feature of one neighbour. We follow the written alignment proof of the repository, whose lemma labels (A-L2.1 and so on) we give with each statement together with the Lean theorem that proves it and that theorem's tier (Table 3); the written proofs are the exposition, and where a formal proof argues differently we say so after the written one.

Fix a feature  $F$  of a tile and write  $E(F)$  for its *closed edge graph*: the four edges of the base square and the four ridges from the base corners to the apex, a compact connected graph without isolated vertices. In tangent coordinates  $u$  on the panel and outward normal coordinate  $z$ , the feature of signed height  $h = a/10000$  is the graph of the tent function  $f_h(u) = h \max(0, 1 - \|u\|_\infty/\eta)$ , and near it the solid is the hypograph  $z \leq f_h(u)$ . Throughout, a *packing* is a family of isometric copies of  $Q$  with pairwise disjoint interiors, and a *tiling* is a packing whose union is  $\mathbb{R}^3$ .

**Lemma 4.1** (local finiteness; A-L2.1, T1 `local_finiteness`). *Every packing by congruent copies of  $Q$  is locally finite.*

*Proof.* Choose a ball of positive radius inside the interior of  $Q$  and carry it along with each tile. The carried balls have pairwise disjoint interiors. A tile meeting a bounded set  $K$  has its ball inside a fixed bounded enlargement of  $K$ , since  $Q$  has bounded diameter, and only finitely many disjoint balls of equal positive volume fit there.  $\square$

**Lemma 4.2** (cone budgets; A-L2.2, written; Lean T1N `cone_sector_budgets_holds`). *At a point common to finitely many tiles of a packing, the interiors of their tangent cones are disjoint, so their solid angles sum to at most  $4\pi$ , with equality in a tiling. At a generic point of an edge, a perpendicular cross-section is partitioned into planar sectors whose angles sum to  $2\pi$  in a tiling.*

*Proof.* Near a boundary point a polyhedral set agrees with a translate of its tangent cone, and one radius serves the finitely many incident tiles; a common open direction would give overlapping interiors. Cone boundaries are finite unions of planar pieces and have spherical area zero. Coverage gives equality in a tiling. The planar statement is the same argument in the perpendicular plane.  $\square$

The word *generic* is made precise as follows. Fix a closed root edge and the finitely many tiles meeting a small neighbourhood of it. Remove their vertices on the edge, their intersections with non-collinear edges, and the intersections with face planes that meet the root line in a single point: finitely many points. At every remaining point an incident tile either has a face containing the line, which contributes a half-plane sector of angle  $\pi$ , or an edge collinear with it, which contributes an ordinary dihedral sector. No tile can contain such a point in its interior, since  $Q$  is the closure of its interior and the root has interior points nearby. Nothing here assumes that a neighbouring facet coincides with a facet of the root; contacts that are not face-to-face are handled by the same sectors.

Put  $t_j = j/100$  for  $j = 1, \dots, 12$ , and

$$\alpha_j = \arctan t_j, \quad \beta_j = \arccos \frac{1}{1 + t_j^2}.$$

**Lemma 4.3** (complete dihedral list; A-L3.1, written; Lean T1N `complete_dihedral_list_holds`; its arithmetic and the mesh audit are separate Lean theorems, see below). *A protrusion of magnitude  $j$  has base dihedral  $\pi + \alpha_j$  and ridge dihedral  $\pi - \beta_j$ ; a recess has the complementary angles. Every other edge of  $Q$  has dihedral  $\pi/2$  or  $3\pi/2$ , and flat triangulation edges have angle  $\pi$ . All 24 deviations  $\alpha_j, \beta_j$  are distinct and less than  $\pi/4$ .*

*Proof.* The facets of the tent have slopes  $(0, 0)$ ,  $(\pm t_j, 0)$  and  $(0, \pm t_j)$ ; their outward normals give the formulas and the signs. The tube construction changes no carrier edge, and the disjoint feature supports create no other kind of edge. Both sequences increase strictly, and a coincidence  $\alpha_i = \beta_j$  would force  $1 + (i/100)^2 = (1 + (j/100)^2)^2$ , that is  $10000 i^2 = 20000 j^2 + j^4$ ; then  $10 \mid j$ , so  $j = 10$  and  $i^2 = 201$ , impossible. The bound  $\pi/4$  follows from  $t_j < 1$  and  $(1 + t_j^2)^2 < 2$  at  $t_{12} = 3/25$ .  $\square$

The unsolvability of  $10000 i^2 = 20000 j^2 + j^4$  over  $1 \leq i, j \leq 12$  is a Lean theorem by kernel `decide` (`deviations_distinct`). The lemma is about the ideal construction; a separate audit, a Lean theorem using `native_decide` (`mesh_angle_audit`), classifies every one of the 6,408 edges of the actual mesh by exact rational outward normals, finds exactly 1,536 feature edges with 32 of each sign for each of the 24 deviations, and checks that every other edge is flat or has orthogonal normals (Table 7).

**Lemma 4.4** (one complementary partner; A-L3.2, written; Lean T1N `generic_feature_partner_holds`). *At a generic point of a feature edge*

*in any tiling exactly two tiles are incident: the root, and one tile whose incident edge has the same type (base or ridge), the same magnitude, and the complementary dihedral angle.*

*Proof.* Every feature sector lies strictly between  $3\pi/4$  and  $5\pi/4$ . Three feature sectors exceed  $2\pi$ , as do two feature sectors together with any ordinary sector, whose angle is at least  $\pi/2$ . A single feature sector is impossible: its angle is  $\pi$  plus a nonzero deviation of magnitude less than  $\pi/4$ , while the remaining sectors are multiples of  $\pi/2$  (a face containing the line contributes  $\pi$ , as noted above), and such a sum is not  $2\pi$ . The root supplies one feature sector, so there are exactly two and nothing else; their deviations are opposite and equal in magnitude, and Lemma 4.3 identifies the type and the magnitude.  $\square$

This does not yet say that an edge angle forces a whole feature. That needs a bound at the vertices of the feature, where sectors are not available.

**Lemma 4.5** (solid-angle bound on the feature graph; A-L4.1, written; Lean T1N `feature_circular_cone_containment_holds`, with the cone formula and its arithmetic T1 `circular_cone_solid_angle_holds`). *At every point of  $E(F)$ , including the apex and the base corners, the interior solid angle of the tile exceeds  $4\pi/3$ ; for  $Q$  it exceeds  $7\pi/4$ .*

*Proof.* The tent is Lipschitz with constant at most  $L = 3/25$ , on its faces, along its creases, and across the transition to the flat panel. At any point of the graph the tile's interior tangent cone contains the circular cone  $\{(v, w) : w < -L\|v\|_2\}$ , whose solid angle is  $\Omega(L) = 2\pi(1 - L/\sqrt{1 + L^2})$ . This exceeds  $4\pi/3$  exactly when  $8L^2 < 1$ , and here  $8L^2 = 72/625$ ; it exceeds  $7\pi/4$  when  $63L^2 < 1$ , and  $63L^2 = 567/625$ .  $\square$

Only the weaker bound is used below: by Lemma 4.2, three distinct tiles cannot share a point lying on a feature graph of each of them.

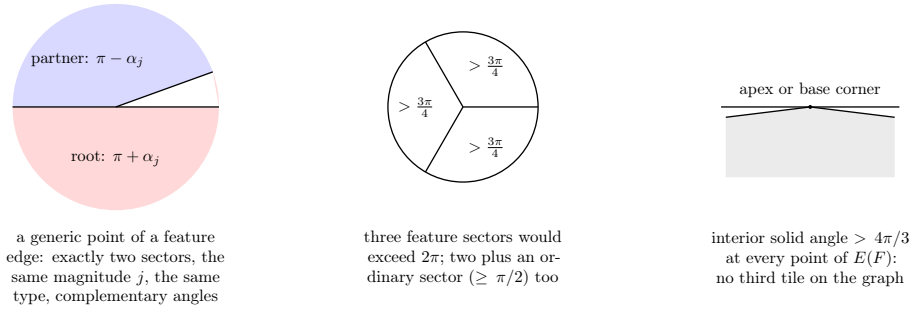


FIGURE 8. The companion argument: sectors at a generic edge point (Lemma 4.4) and the solid-angle cone at a vertex of the feature graph (Lemma 4.5).

**Theorem 4.6** (one tile accompanies the whole feature graph; A-T4.2, written; Lean T1N `connected_feature_companion_holds`). *For each feature  $F$  of a tile  $R$  in any tiling there is a unique other tile  $S$  such that  $E(F)$  is contained in the union of the feature edge graphs of  $S$ ; moreover  $E(F)$  lies in one feature edge graph of  $S$ .*

*Proof.* By Lemma 4.1, only finitely many tiles other than  $R$  meet  $E(F)$ . For each such tile  $S \neq R$ , let  $C_S$  be the intersection of  $E(F)$  with the union of the feature edge graphs of  $S$ ; each  $C_S$  is closed in  $E(F)$ . The sets  $C_S$  cover  $E(F)$ : the generic interior points of the root's edges are covered by Lemma 4.4, they are dense in  $E(F)$ , and a finite union of closed sets containing a dense subset is everything. The sets  $C_S$  are pairwise disjoint: a point in  $C_S \cap C_T$  with  $S \neq T$  would lie on a feature graph of  $R$ , of  $S$  and of  $T$ , which Lemmas 4.2 and 4.5 forbid. A connected space is not the union of two or more nonempty members of a finite disjoint closed cover, each member being also open. Hence exactly one  $C_S$  is nonempty and equals  $E(F)$ . The feature graphs of  $S$  are disjoint compact sets, and the same connectedness argument applied to  $E(F)$  inside their union puts  $E(F)$  in one of them.  $\square$

This single argument disposes of every case in which a partner might change: at an interior edge point, at a subdivision vertex of the mesh, at the apex, at a base corner, or between two features of the same neighbour. It needs no face-to-face hypothesis and no continuation rule supplied by the observer.

**Lemma 4.7** (containment implies equality; A-L4.3, written, with erratum E1; Lean T1N `feature_containment_rigidity_holds`). *The features  $F$  and  $F'$  of Theorem 4.6 have the same closed edge graph, the same magnitude, and opposite polarity; their triangulated surfaces coincide.*

*Proof.* Choose a point in the interior of a base edge of  $F$  that avoids the vertices of the companion's graph and the finite exceptional set of Lemma 4.2; such a point exists. There Lemma 4.4 gives  $F'$  the same magnitude  $j$  and the same edge type. A feature of magnitude  $j$  has total edge length  $\ell_j = 8\eta + 4\sqrt{2\eta^2 + (j/10000)^2}$ . We know  $E(F)$  is a closed subset of  $E(F')$ ; if the inclusion were proper, a point of  $E(F')$  outside  $E(F)$  would have positive distance from the closed set  $E(F)$ , and since  $E(F')$  has no isolated vertices it would contribute a nonzero subsegment outside  $E(F)$ , giving  $\ell_j < \ell_j$  by additivity of length. So the graphs are equal. In the graph the apex is the unique vertex of degree four and the base corners have degree three, so equality recovers the base square, the apex, and the four triangles; complementary base dihedrals force opposite polarity.  $\square$

The formal proof argues differently: the graph  $E(F)$  is compact and the placement carrying it into  $E(F')$  is an isometry, and an isometric self-map of a compact metric space is onto, so the containment is an equality without the length count; the premise that the two tiles are distinct is used there.

**Corollary 4.8** (a feature match discretizes the pose; A-C4.4, written; Lean `T1N_companion_pose_discrete_holds`). *Normalize the root's pose to  $(I, 0)$ . Every feature companion has a signed permutation matrix  $G$  as linear part and translation  $t = c_u - Gc_v \in \frac{1}{8}\mathbb{Z}^3$ , where  $c_u, c_v$  are native feature centres with  $a_u = -a_v$  and outward base normals  $n_u = -Gn_v$ .*

*Proof.* A rigid motion matching two complete pyramid graphs carries base-edge directions and base-plane normal to each other, and those are the coordinate directions, so its orthogonal part is a signed permutation, of either determinant. The graph identifies the centre; opposite polarity and a common apex force opposite outward normals; all centres lie in  $\frac{1}{8}\mathbb{Z}^3$ .  $\square$

No global alignment has been concluded yet: only the poses of complete feature companions have been reduced to a finite list, which the next section examines exhaustively. The mechanism is the one Socolar and Taylor use for their three-dimensional tile, where plugs enforce a matching rule by shape [39]; here the pyramids enforce, by dihedral angles and solid angles alone, that every feature is matched by a feature of the opposite polarity on one neighbour.

## 5. EVERY TILING IS REGISTERED

Let  $F_0$  be the finite set of poses given by Corollary 4.8: for every ordered pair of native features of opposite signed height and every signed permutation  $G$  with  $n_u = -Gn_v$ , the pose  $(G, c_u - Gc_v)$ . There are 6,862 of them, with translations in eighth units, and for each pose we retain the list of root features it can fully accompany. Each is a candidate for the pose of a tile sharing a feature with the root; the task is to show that only 44 of them, all integral and proper, can occur in a tiling.

**Lemma 5.1** (eighth-grid overlaps penetrate the cores; A-L5.1; the overlap implication is the Lean theorem `retained_core_overlap_holds` (T1), the quantitative bound the written estimate in the proof). *If two poses in  $F_0$  have baseline chairs overlapping in positive volume, the corresponding copies of  $Q$  overlap in positive volume; more precisely, an overlapping pair of unit cubes yields an open box of side at least  $21/200$  inside both interiors, hence a ball of radius  $21/400$ .*

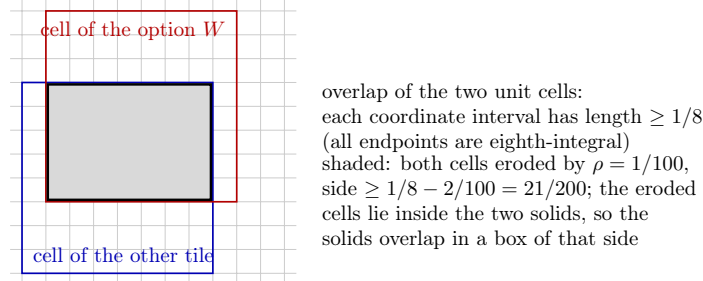
*Proof.* Cube endpoints are eighth-integral, so each positive coordinate overlap has length at least  $1/8$ ; eroding both cubes by  $\rho = 1/100$  leaves at least  $1/8 - 2/100 = 21/200$ , and the eroded cubes lie in the interiors of the solids (Section E.5).  $\square$

Let  $F_1$  be the poses in  $F_0$  whose baseline chair does not overlap the root's, and for a root feature  $u$  let  $\text{Partners}(u)$  be *all* poses in  $F_1$  matching  $u$ ; no other rejection enters these lists. The census is summarized in Table 6 and Section B: of the 6,862 poses, 1,545 collide with the root directly, leaving 5,317 in  $F_1$ , of which 5,234 have a non-integral translation and 83 are integral.

The remaining step rejects 5,273 of the 5,317 by the following finite statement, leaving 44.

**Proposition 5.2** (collision witnesses; A-FS5.2, T1N `mates_census`, and two independent replays). *For each of the 5,273 rejected poses  $V \in F_1$  the certificate names a feature  $u$  of either the root  $U$  or of  $V$ , and, in its owner’s coordinates: (i) the other tile of the pair is not in  $\text{Partners}(u)$ ; (ii)  $\text{Partners}(u)$  is nonempty; (iii) every  $W \in \text{Partners}(u)$  has a positive-volume baseline overlap with the other tile of the pair. There are 299,975 such option collisions in all.*

The verifier regenerates  $F_0$  and every  $\text{Partners}(u)$  from the actual mesh and checks the three clauses; it uses no flat-panel consistency filter and no already-rejected pose when building the option sets, so the conclusion is not used to prune its own options. For every option it exports one pair of overlapping native cubes with their intersection box, and a second, integer-only checker verifies all 299,975 boxes in units of  $1/400$ ; every box has sides at least  $42/400$  (Figure 9).



Schematic coordinate-plane section of the overlap estimate  
(the certificate stores the actual boxes in integer units of  $1/400$ )

FIGURE 9. Schematic of the retained-core overlap estimate used in Proposition 5.2; this is not a selected certificate witness.

**Lemma 5.3** (only the 44 survivors occur; A-L5.3, written; Lean T1 `only_registered_mates_holds`). *No rejected pose is a feature companion in any tiling. The 44 surviving poses are integral and proper, and they are exactly the atlas  $\mathcal{A}_{44}$  of Section 3.1.*

*Proof.* Suppose tiles  $U$  and  $V$  occur in a tiling in a rejected relative pose. The named feature  $u$  has a complete companion  $W$  by Theorem 4.6 and Lemma 4.7, whose pose lies in  $\text{Partners}(u)$  because the list of complete-feature poses is exhaustive (Corollary 4.8) and an overlap with the owner is forbidden; by clause (i)  $W$  is not the other tile already placed. Clause (iii) says  $W$ ’s baseline overlaps that other tile’s, and Lemma 5.1 turns this into an overlap of interiors, a contradiction. The verifier compares the set of 44 survivors,

not merely its size, with the atlas computed from the profile in Section 3.1; they are equal.  $\square$

This is a bounded local obstruction with a complete geometric carrier theorem behind it. It is not a search for periodic cells, not a timeout, not a sample of surface points, and not an inference that a partial gap between misaligned tiles cannot be filled by a third tile: every possible filler of the missing companion is exhibited as colliding with a whole tile.

**Lemma 5.4** (a component’s baseline cells exhaust the grid; A-L6.1, written; Lean T1N `baseline_component_covers_grid_holds`). *Choose any root tile in a tiling and normalize its pose to  $(I, 0)$ . In the graph whose edges join tiles sharing a complete feature companion, let  $\mathcal{C}$  be the root’s component. Every tile in  $\mathcal{C}$  has an integer translation and a proper cubic frame, and the baseline chairs of  $\mathcal{C}$  tile the lattice of unit cubes.*

*Proof.* Every edge of the graph has a relative pose in  $\mathcal{A}_{44}$  (Lemma 5.3), and compositions preserve integrality and properness. Two baseline chairs of  $\mathcal{C}$  cannot share a cell, since their retained cores would overlap. Let  $W$  be the set of unit cells covered by the chairs of  $\mathcal{C}$ . Take a cell  $a \in W$  and a face-adjacent cell  $b$ . If  $b$  lies in the same chair it is in  $W$ . Otherwise the common face is an exposed panel of  $a$ ’s chair, carrying eight features; the companion of any one of them lies in  $\mathcal{C}$  with an integral pose, its base normal is opposite, and its centre coincides with the root feature’s centre at a point strictly inside an integral unit panel; two integral unit squares in that plane whose relative interiors share the point are the same square, so the companion owns  $b$ . Thus  $W$  is closed under lattice steps, and  $W = \mathbb{Z}^3$ , with exactly one owner per cell.  $\square$

**Lemma 5.5** (the component’s solids cover  $\mathbb{R}^3$ ; A-L6.2, written, with erratum E6; Lean T1 `component_solids_cover_holds`). *The copies of  $Q$  in  $\mathcal{C}$  cover  $\mathbb{R}^3$  with disjoint interiors.*

*Proof.* By Lemma 5.4, across every panel shared by two chairs of  $\mathcal{C}$  the adjacent cell has one owner, and each of the eight features on the panel has its companion in that owner, since another owner would duplicate the cell; so the two profiles agree on the whole panel. Around each paired feature take its closed normal tube of half-height  $\rho$  and apply the tube map of Section E.5; opposite polarity and opposite outward normals give the same map from either side, so this is one map, not two. The tubes are disjoint except for coincident pairs and are locally finite; the maps are the identity outside the tubes and on their boundaries, and glue with their inverses to a homeomorphism  $H$  of  $\mathbb{R}^3$ . Every feature tube belongs to the two cells on either side of its panel, both with unique owners in  $\mathcal{C}$ , and a tube not on a tile’s exposed boundary is disjoint from that tile’s material; hence  $H$  carries each baseline chair  $GP + t$  of  $\mathcal{C}$  exactly onto  $GQ + t$ , and the images cover  $\mathbb{R}^3$  with disjoint interiors.  $\square$

The formal proof avoids the assembly-wide gluing. For any tile of the given tiling, choose a component carrier containing the centre of its inscribed quarter-ball. Clamping produces an interior point of the component's physical tile at squared distance at most  $3/2500 < 1/100$  from that centre, hence inside the quarter-ball. Packing disjointness forces the two tiles to coincide. Thus every tile belongs to the component, and the original tiling's coverage gives component coverage. This uses an existing tiling and is not an existence proof.

**Theorem 5.6** (unrestricted alignment; A-T6.3, written; assembled in Lean as `registration_of_tiling` from `unrestricted_alignment_holds`, T1N). *Every tiling by  $Q$  is registered: there is one ambient isometry after which every tile has a proper cubic frame and an integer translation, and every pair of tiles sharing a feature is in a relative pose of  $\mathcal{A}_{44}$ .*

*Proof.* The component  $\mathcal{C}$  of Lemma 5.5 already fills space. A tile outside  $\mathcal{C}$  would have nonempty interior, while the boundaries of the tiles of  $\mathcal{C}$  form a locally finite union of polyhedral surfaces containing no ball; an interior point of the extra tile would then lie inside a tile of  $\mathcal{C}$ , contradicting disjointness of interiors. So there is no other component, and the ambient isometry that normalizes the root registers every tile.  $\square$

Registration is one ambient isometry applied to the whole tiling, so it preserves periodicity in the Spectre paper's sense [37]: the registered tiling has the conjugate symmetry group, and if the ambient isometry has linear part  $A$ , a period  $p$  of the original tiling becomes the period  $Ap$  of the registered tiling, with  $\|Ap\| = \|p\|$ . This closes the loophole of an integral network of tiles surrounded by tilted or shifted tiles in residual gaps: the tube homeomorphism proves physical coverage, not merely the occupancy of a grid. The relative poses in  $\mathcal{A}_{44}$  are all proper, although all 48 signed frames were admitted as candidates; a tiling may be reflected as a whole, but its two handedness sectors never mix, which is the content of Theorem 1.2(3) proved in Section 7. In Lean, a registration is a record of an ambient isometry and, for each placement, a frame index among the 24 and an integer shift; the theorem that every tiling has one is assembled from the three written lemmas of this section, which appear by name in Section C.

## 6. THE UNIQUE HIERARCHY

From now on every tiling is registered (Theorem 5.6), and the argument is finite combinatorics on the lattice plus one scale argument. For each of the eight children of Table 1 there is one candidate parent through a root chair in the identity pose: apply the inverse child pose to the list of children. Each candidate is a cluster of eight chairs whose baseline is a congruent copy of  $2P$ .

**Lemma 6.1** (first shells; R§7, T1N `first_shells_33`). *There are exactly 33 ways to cover the 22 shell cells of a root chair by legal neighbours from  $\mathcal{A}_{44}$*

*without overlaps and without an illegal contact between chosen neighbours. Each of the 33 contains the face-adjacent sibling signature of exactly one candidate parent.*

The 33 arrangements (Figure 10) are necessary local configurations, not claims that each extends to a tiling; a second implementation, using a set-based recursion instead of the generator’s bitmask recursion, reproduces all 33.

**Lemma 6.2** (central completion; R§7, T1N `central_completion`). *In one of the 33 arrangements the root is the central child of its proposed parent and all seven outer children are present. In the other 32 the proposed parent’s centre is a neighbour of the root; checking every first shell at that centre for compatibility with the arrangement, 18 of the 32 have no compatible completion and cannot occur in a tiling, 14 have completions, and every completion makes the proposed centre genuinely central.*

**Lemma 6.3** (parent conflicts; R§7, T1N `parent_conflicts_28`). *Each of the 28 unordered pairs of distinct candidate parents through one root contains an incompatible pair of child poses.*

**Theorem 6.4** (unique parent; R§7 Theorem 7.1; `parent_local_to_global`, `unique_parent`: T1N, unconditional). *Every tiling by  $Q$  partitions uniquely into the eight-chair parent clusters, and the parent of each tile is determined by a finite neighbourhood of it.*

*Proof.* Registration gives the complete local inventory around a tile: its first shell is one of the 33 of Lemma 6.1, which proposes a unique parent. By Lemma 6.2 that parent’s seven other children are present in every tiling in which the shell occurs. By Lemma 6.3 two parents cannot share a tile, so the local assignments are globally disjoint and consistent, and every tile is included.  $\square$

**Theorem 6.5** (admissibility-preserving coarsening; R§8 Theorem 8.1; T1N, unconditional). *Let  $D$  group a tiling  $T$  into its unique parents, divide the parent origins and the scale by two, and replace each parent’s baseline by a fresh copy of  $Q$  in that frame. Then  $D(T)$  is again a tiling by  $Q$ , and  $D(T + v) = D(T) + v/2$  for every vector  $v$ .*

In Lean the two clauses are `coarsening_is_tiling` and `coarsening_translation_equivariant`; the physical realization of the coarsened tiling uses the theorem `small_collar_realization_holds` (T1N, Section 3.4) and the tube construction.

*Proof.* The parent bodies partition space and their contact graph is connected, since the unit-cell adjacency graph is. Every contact between two parents is one of the 44 legal parent contacts of Section 3.2, whose translations are all even, so all parent origins lie in one coset of  $2\mathbb{Z}^3$  and halving puts them on a common lattice; by the identity of Section 3.2 every halved contact belongs

| shell | atlas contacts        | parent | completion  |
|-------|-----------------------|--------|-------------|
| 1     | 3,9,10,12,20,29,38,40 | 7      | central     |
| 2     | 10,13,16,18,21,22,41  | 0      | completable |
| 3     | 1,9,10,13,22,27,37,41 | 0      | completable |
| 4     | 2,4,11,16,18,21,42    | 6      | completable |
| 5     | 4,11,14,16,18,21,42   | 6      | impossible  |
| 6     | 11,14,15,16,18,21,42  | 2      | completable |
| 7     | 4,11,16,18,21,22,42   | 6      | impossible  |
| 8     | 11,16,18,21,22,23,42  | 1      | completable |
| 9     | 11,14,16,18,21,26,42  | 3      | impossible  |
| 10    | 11,16,18,21,22,26,42  | 3      | impossible  |
| 11    | 11,16,18,21,25,26,42  | 3      | completable |
| 12    | 11,14,16,18,21,34,42  | 5      | impossible  |
| 13    | 11,16,18,21,22,34,42  | 5      | impossible  |
| 14    | 11,16,18,21,33,34,42  | 5      | completable |
| 15    | 1,2,4,9,11,27,37,42   | 6      | completable |
| 16    | 1,4,9,11,14,27,37,42  | 6      | impossible  |
| 17    | 1,9,11,14,15,27,37,42 | 2      | completable |
| <hr/> |                       |        |             |
| shell | atlas contacts        | parent | completion  |
| 18    | 1,4,9,11,22,27,37,42  | 6      | impossible  |
| 19    | 1,9,11,22,23,27,37,42 | 1      | completable |
| 20    | 1,9,11,14,26,27,37,42 | 3      | impossible  |
| 21    | 1,9,11,22,26,27,37,42 | 3      | impossible  |
| 22    | 1,9,11,25,26,27,37,42 | 3      | completable |
| 23    | 1,9,11,14,27,34,37,42 | 5      | impossible  |
| 24    | 1,9,11,22,27,34,37,42 | 5      | impossible  |
| 25    | 1,9,11,27,33,34,37,42 | 5      | completable |
| 26    | 4,11,16,18,21,42,43   | 6      | impossible  |
| 27    | 11,16,18,21,26,42,43  | 3      | impossible  |
| 28    | 11,16,18,21,34,42,43  | 5      | impossible  |
| 29    | 1,4,9,11,27,37,42,43  | 6      | impossible  |
| 30    | 1,9,11,26,27,37,42,43 | 3      | impossible  |
| 31    | 1,9,11,27,34,37,42,43 | 5      | impossible  |
| 32    | 11,16,18,21,42,43,44  | 4      | completable |
| 33    | 1,9,11,27,37,42,43,44 | 4      | completable |

FIGURE 10. The 33 first shells, their proposed parent types and central-completion outcomes.

to  $\mathcal{A}_{44}$ . The halved baseline tiling is therefore a registered chair tiling whose profiles match on every shared panel, and the small-collar realization lemma (Section 3.4) turns it into a tiling by  $Q$ . Translation covariance holds because the parents of  $T + v$  are the parents of  $T$  translated by  $v$  (Theorem 6.4), and the origins are then divided by two; no phase is chosen, the actual parent origins are used.  $\square$

The coset of  $2\mathbb{Z}^3$  may be  $(1, 1, 1) + 2\mathbb{Z}^3$ , so that  $D(T)$  sits on a half-integer coset after halving; periods are differences of poses within one coset and remain integer vectors, which is all Section 7 uses.  $D$  is not a homothety

of each cluster’s corrugated surface into  $Q$ : it decodes the carrier and re-standardizes the surface by the same admissible rule, which is why an exact self-similar corrugated macro-shape is not needed.

Iterating  $D$  gives the hierarchy. Call a *supertile of level  $n$*  a  $2^n$ -scaled chair in a registered pose, and say that a family of partitions of the registered cells of  $T$  into level- $n$  supertiles, one partition for each  $n$ , is *nested* if every level- $(n+1)$  supertile is exactly the union of its eight geometric child supertiles at level  $n$ .

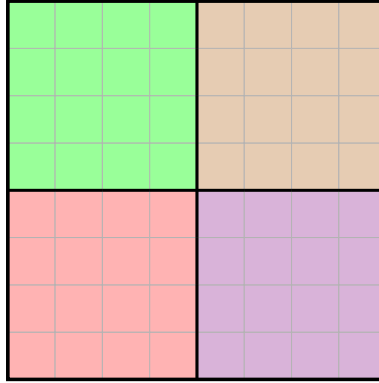
**Theorem 6.6** (the intrinsic hierarchy; ruling R7; T1N, unconditional: `carrier_hierarchy`, `carrier_hierarchy_unique`, `geometric_hierarchy_canonical`, `geometric_hierarchy_unique`). *Every tiling  $T$  by  $Q$  carries a nested family of partitions into supertiles of every level, with level 0 the registered carrier tiling associated with  $T$ , and any two nested families coincide level by level.*

*Proof.* Existence: the level- $n$  partition is read off  $D^n(T)$ , each level- $n$  supertile being a chair of  $D^n(T)$  scaled by  $2^n$  back to the coordinates of  $T$ ; nesting is the definition of  $D$ . Uniqueness: a nested family’s level- $(n+1)$  partition is a partition of the level- $n$  supertiles into parents, which Theorem 6.4 applied to  $D^n(T)$  determines; induct on  $n$ .  $\square$

The pinned Lean declarations prove this in two steps. `carrier_hierarchy_unique` is stated in the type whose level- $n$  poses belong to  $D^n(T)$  (the structure `LevelSupertilePose` carries that membership). The structure `GeometricHierarchy` carries no such membership: its data are the raw poses, each level registered by one common rigid motion, with the nesting law of the theorem; `geometric_hierarchy_canonical` proves, by induction on the level from the unique-parent theorem, that every pose of such a family lies in  $D^n(T)$ , and `geometric_hierarchy_unique` then gives uniqueness among arbitrary geometrically nested registered families, `carrierHierarchy_geometric` supplying the existence half in the raw type. The written induction above is the content of the first of these theorems. The raw structure has no canonicity-membership field. The must-fail control (`negative/NoncanonicalGeometricHierarchy.lean`) attempts to prove that a level-one pose is noncanonical using `geometric_hierarchy_canonical`; it checks that the positive membership proof is rejected at the negated type.

Two things are deliberately not claimed. The theorem has no exhaustion clause: it does not say that the supertiles containing one fixed tile grow to cover space, and they need not. The substitution fixed point with seed  $(6, 0, 4, 5, 2, 3, 0, 1)$  (Section 8.2) has near the origin two whole tiles in the frame  $A = \text{diag}(-1, -1, 1)$ , at the origin and at  $(1, 1, -1)$ ; the first is its own child 000 at every level, so its ancestors are the supertiles  $2^n AP$ , whose union is the closed octant  $\{x \leq 0, y \leq 0, z \geq 0\}$  and not all of space, and the second is its own central child, its ancestors filling the closure of the complement. This is a tiling with an infinite fault surface on supertile boundaries at all levels, of the kind the hat paper contemplates for hats: “it is conceivable that

[tiles] could tile infinite sectors of the plane, which could then be combined into tilings with infinite ‘fault lines’ that lie on the boundaries of supertiles at all levels” [36]. It is nevertheless substitution-legal, every finite patch of it occurring inside a supertile, so non-exhaustion does not by itself take a tiling out of the substitution hull; whether every tiling by  $Q$  lies in that hull is a separate question, left open (Section 10). Nor is the hierarchy claimed to be locally derivable from the tiling by unmarked chairs: an unmarked chair has eight proper orientations while a registered pose of  $Q$  is one of 24, so the frame of a chair carries a  $\mathbb{Z}/3$  of information invisible to its geometry, and the parent is read from the features, not from the chairs. What the theorem gives is exactly what Section 7 needs: a grouping at every scale that the tiling determines and that every symmetry of the tiling therefore preserves. The same-frame grandchild at  $(2, 2, 2)$  and the first two nested patches (64 and 4,096 tiles, 448 and 28,672 cells) are checked in Lean as controls (Figure 11).



slice  $0 \leq z < 1$  of the 64-chair patch  $4P$  (side 8):  
cells coloured by level-1 supertile; thick lines: level-1 outlines, level-2 outline

FIGURE 11. The  $0 \leq z < 1$  unit-cell slice of the 64-chair patch  $4P$ ; colours identify level-1 parents, and bold lines show level-1 and level-2 boundaries.

## 7. APERIODICITY AND HOMOCIRALITY

**Theorem 7.1** (no periods; R§10; T1N, unconditional: `r44_einstein`, using `period_halving` and `all_periods_grid`; terminal arithmetic lemma `no_period`: T1). *Every tiling  $T$  by  $Q$  has  $\text{Per}(T) = \{0\}$ .*

*Proof.* Let  $p$  be a translational period of  $T$ . Registration (Theorem 5.6) and the absence of nonidentity self-isometries (Section E.3) make  $p$  an integer vector: the translation by  $p$  carries a registered tile to a registered tile in the same frame, and the difference of two integer shifts is an integer vector. The parent partition is translation-covariant (Theorem 6.4), so  $D(T + p) = D(T) + p/2$  by Theorem 6.5; since  $T + p = T$ ,  $p/2$  is a period of

$D(T)$ , which is again a registered tiling by  $Q$ . Iterating,  $p/2^n$  is a period of  $D^n(T)$  and hence an integer vector for every  $n$ , so  $p \in \bigcap_n 2^n \mathbb{Z}^3 = \{0\}$ .  $\square$

This is the classical argument that a unique hierarchy forbids periods, in the form the Spectre paper states [37] and in the form Robinson's tilings first exhibited: a period would have to be a period of every level, and the levels grow without bound.

**Theorem 7.2** (finite symmetry; R§10; T1N, unconditional: r44\_einstein; conditional bound sym\_card\_le\_24: T1). *Every tiling  $T$  by  $Q$  has  $|\text{Sym}(T)| \leq 24$ .*

*Proof.* After registration every tile's frame lies in one coset of the 24-element proper cubic group, and  $Q$  has no self-isometry, so a symmetry of  $T$  is determined by where it sends one chosen tile: its linear part is one of at most 24 frames, and, that being fixed, its translation part is determined as well. The map from  $\text{Sym}(T)$  to linear parts has kernel the translational symmetries, which are trivial by Theorem 7.1; hence it is injective into a set of at most 24 elements.  $\square$

**Corollary 7.3** (strong aperiodicity; a deduction from the two theorems above). *No tiling by  $Q$  has a symmetry of infinite order. Hence  $Q$  is a strongly aperiodic monotile of  $\mathbb{R}^3$  in the sense of Mozes and of the hat paper, and Theorem 1.1 holds.*

*Proof.* A finite group has no element of infinite order; existence is Theorem 3.1.  $\square$

In the taxonomy of Coulbois, Gajardo, Guillon and Lutfalla, Theorem 7.2 says that  $Q$  is mildly aperiodic. Strong aperiodicity in their sense would require every tiling to have trivial symmetry group, and  $Q$  does not have that property: the substitution fixed points of Proposition 8.2 include tilings whose symmetry group has order 8, as well as tilings with trivial symmetry group. In particular no screw motion, rational or irrational, is a symmetry of any tiling by  $Q$ , which is the property the biprism lacks.

**Theorem 7.4** (homochirality; tiling\_homochiral, tiling\_chirality\_corollary: T1N, unconditional). *In every tiling  $T$  by  $Q$  all placements have the same handedness: the determinants of their linear parts are all  $+1$  or all  $-1$ . Consequently a reflected and an unreflected copy of  $Q$  never occur in the same tiling, and every tiling is homochiral.*

*Proof.* After the ambient isometry of Theorem 5.6, every placement is a proper cubic frame followed by a translation, so all placements have determinant  $+1$  relative to that isometry; undoing it multiplies all determinants by the same sign. For two tiles  $gQ$  and  $hQ$  the relative motion  $hg^{-1}$  then has determinant  $+1$ , which is the Spectre paper's definition of a homochiral tiling.  $\square$

**Corollary 7.5** (strictly chiral aperiodic monotile; a deduction at the same verification boundary).  *$Q$  is a strictly chiral aperiodic monotile of  $\mathbb{R}^3$  in the*

*sense of the Spectre paper: it admits tilings, and every tiling it admits is homochiral and non-periodic.*

Reflections were never excluded by convention: all 48 signed frames were admitted as candidate poses in Sections 4 and 5, and the 44 legal contacts turned out proper. A tiling may be reflected as a whole; what the geometry forbids is a mixture. The Schmitt–Conway–Danzer biprism, by contrast, needs reflections forbidden for its aperiodicity [40]. The four clauses of Theorem 1.2 are now proved: (1) is Theorem 3.1, (2) is Theorems 7.1 and 7.2, (3) is Theorem 7.4, and (4) is Theorem 6.6.

## 8. MECHANIZATION AND THE LIMIT-PERIODIC COROLLARY

**8.1. What is checked by machine.** The finite parts of the proof are theorems of a Lean 4 development written for this paper, with Mathlib as its library, and the logical assembly of the whole argument is a Lean theorem as well. Table 2 lists the 23 finite theorems with their statements and the method by which each is decided: by the kernel’s evaluator (`decide`) where that is feasible, and otherwise by `native_decide`, which compiles the decision procedure and trusts its result through one named axiom per theorem. Those axioms are the only non-standard ones in the development: Table 4 gives, for every finite theorem and for the main theorems, the output of Lean’s `#print axioms`, which for the main theorem consists of the three standard axioms of classical Mathlib and the 21 named hooks of the finite checks it uses. No theorem in the development depends on `sorryAx`, and no admission remains.

The assembly is the theorem `r44_einstein`, which takes no hypothesis: it states that a tiling exists, that every tiling has  $\text{Per}(T) = \{0\}$ , and that  $|\text{Sym}(T)| \leq 24$ ; the theorems `tiling_homochiral` and `carrier_hierarchy`, with `geometric_hierarchy_unique`, state clauses (3) and (4) of Theorem 1.2. The development was written with a record `Hypotheses` of named written lemmas as its verification boundary, and the written lemmas of this paper are the fields that record had: the tube construction and carrier recovery in Sections E.3 and E.5, realization in Section 3.4, and the companion and registration arguments in Sections 4 and 5. Every one of them is now a Lean theorem of the pinned development (Table 3 lists the 19 theorems, with their tiers by their own axiom lines), the record is empty, and the conditional form survives under the name `r44_einstein_of_hypotheses` over the empty record (Section C). Everything else, including the period-halving tower, the symmetry bound, local finiteness, the unique parent, the coarsening and the hierarchy with its uniqueness among arbitrary geometrically nested registered families (Theorem 6.6), is proved inside Lean. The honest summary is therefore this: the theorem is kernel-checked modulo the named compiler hooks; the written proofs of Sections 3.4 and E and Sections 4 and 5 remain as exposition, independently reviewed, and four of the formal proofs argue differently from the written ones, as noted where those lemmas appear (carrier recovery, containment rigidity, the covering by a component’s solids, and

small-collar realization). The same pipeline was exercised on the hat as a calibration of the tooling (Section D). The hat and Spectre papers rely on no proof assistant; the nearest comparator is Myers’s in-progress formalization of those papers [28], and we claim no coverage by Mathlib itself [42].

Independently of Lean, every enumeration is replayed from the Python standard library by one command in about a minute (Section A), and the companion census was checked by two independent Python implementations; the second extracts the features from the actual triangulated mesh rather than from their annotation table. No floating-point value enters any checked statement: certificates are integers and exact rationals, the checkers use exact arithmetic, and the Lean theorems decide integer literals. Negative controls accompany the positive build. The diagonal audit distinguishes assertions about distinct companion tiles from relations with legitimate reflexive cases; the controls exercise the repaired diagonal application, corrupted finite inputs, mixed-handedness claims, a second-hierarchy claim and a claim that a raw geometric hierarchy has a noncanonical level-one pose, each checked to fail with its expected diagnostic. Six corrupted inputs to the replay are rejected. The development was reviewed adversarially twice before this paper was written; the first round found the hypothesis record inconsistent (one field admitted comparing a tile with itself), which was repaired and controlled, and the second round found no blocking or major issue, while noting that the reviewer could not run Lean. A third round, on the closed record with the empty hypothesis record, was requested on 2026-09-10; as of that date its verdict is pending. We make no claim of simplicity, and we have one route to the theorem: the continuous argument of Section 4 is written for checking by hand with its arithmetic in Lean, and the finite part is where the computer does the work.

*Code.* The repository containing the solid, the certificates, the replay, the Lean development and the viewer accompanies this paper (Section A); its mathematical baseline is commit `d90313a717`, the intended release tag is `paper-v1`, and tag creation and licensing remain pending owner actions.

*Version freeze.* This review version uses commit `d90313a717` (2026-09-10) as its mathematical and Lean baseline. The 19 theorems of Table 3, the empty record, and the recorded axiom dependencies of Table 4 refer to that baseline; the geometric certificate data and Lean statements refer to that baseline, and the bibliography and replay measurements have the separately identified sources recorded in the paper. The intended paper-release tag is `paper-v1`, to be placed on the paper-release commit; tag creation and licensing remain pending owner actions. The theorem, the written proof and the finite certificates are unchanged since the earlier baseline `dd2735d9bd` (2026-09-08), at which 16 of the 19 written lemmas were still hypotheses of the Lean assembly.

*Evidence, not proof.* Independently of the proof, and used nowhere in it, a theorem-free search was run over grid-registered copies of  $Q$  (all 48 signed

frames, reflections included; copies compatible when their carrier cells are disjoint and the features on every shared unit panel are exact opposites, in rational arithmetic; `simulations/periodicity\_search.py`). A SAT solver was asked, for every full-rank sublattice  $L \subset \mathbb{Z}^3$  of index at most 40, whether the torus  $\mathbb{R}^3/L$  can be tiled by such copies; all 43,981 instances are unsatisfiable, so no tiling by registered copies has a full-rank period lattice of index at most 40. Heesch-style coronas were grown around one copy to radius 5 of 5 tried, with 251 copies at radius 5. The positive controls of the same script find the unit cube periodic at index 1 and the featureless chair carrier periodic at index 7, as recorded in `simulations/review/2026-09-09\_periodicity\_search\_r1\_eddy.md`. The torus and corona receipt is `simulations/periodicity\_search\_report.json` (wall time 108,682 s). This is evidence in the sense of a sanity check on the finite data: it concerns registered copies and full-rank lattices only, whereas Theorem 7.1 excludes every period of every tiling.

TABLE 2. The finite theorems of the Lean development at the state described in this paper: statement in words (the theorem’s docstring) and decision method (FIGURES.md, Table 1).

| theorem                            | statement (docstring)                                                                                                                                                                                                                                                   | method                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <code>children_partition_2P</code> | The eight literal child poses are proper and their 56 integer cells are a disjoint, exact partition of the doubled seven-cube chair.                                                                                                                                    | <code>kernel decide</code> |
| <code>contact_closure_30</code>    | Refining the 21 internal face contacts once gives 30 states, and one more refinement adds no state. The result is the literal 30-state certificate.                                                                                                                     | <code>native_decide</code> |
| <code>profile_canonical</code>     | The 30 states induce 372 distinct sign equations. Their listed twelve connected components are balanced, rooted positively at their least roles, and ordered by those roots; canonical signs and component magnitudes give both independent 192-entry profile literals. | <code>native_decide</code> |
| <code>atlas_44</code>              | Among all 2,388 disjoint touching shell poses in all 48 signed frames, profile compatibility admits exactly the literal 44-pose atlas.                                                                                                                                  | <code>native_decide</code> |
| <code>first_shells_33</code>       | Exhaustive exact-cover recursion gives precisely the 33 literal covers of the 22-cell shell, and every cover has one candidate parent signature.                                                                                                                        | <code>native_decide</code> |

| theorem                             | statement (docstring)                                                                                                                                                                                                                                                                                        | method               |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>central_completion</b>           | Of the 32 outer-root shells, exactly 14 have a compatible completion and 18 have none; every viable completing shell types the centre as central.                                                                                                                                                            | <b>native_decide</b> |
| <b>parent_conflicts_28</b>          | Every one of the 28 unordered pairs among the eight parents through a root contains an incompatible pair of child poses.                                                                                                                                                                                     | <b>native_decide</b> |
| <b>parent_atlas_eq_fine</b>         | The induced parent census is 697 candidates, 116 disjoint macro-bodies and 44 admitted contacts; all admitted shifts are even and halving gives the fine 44-contact atlas extensionally.                                                                                                                     | <b>native_decide</b> |
| <b>mates_census</b>                 | The literal complete-feature census has 6,862 poses, of which 5,317 avoid the root baseline. The 5,273 rejection records quantify over every isolated companion option (299,975 in all); every selected overlap retains side at least $42/400 = 21/200$ . The 44 survivors are exactly the registered atlas. | <b>native_decide</b> |
| <b>deviations_distinct</b>          | For $1 \leq i, j \leq 12$ the base/ridge deviation equation has no solution. For $L=3/25$ , both requested rational inequalities hold after clearing $25^2$ .                                                                                                                                                | kernel decide        |
| <b>no_native_symmetry</b>           | Among all 48 signed coordinate permutations, only the identity preserves the centres, outward normals and coefficients of the 192-feature table.                                                                                                                                                             | kernel decide        |
| <b>native_features_length</b>       | The literal table has exactly the 192 roles used by the geometric model.                                                                                                                                                                                                                                     | kernel decide        |
| <b>native_feature_centers_nodup</b> | Distinct literal roles have distinct eighth-grid feature centres.                                                                                                                                                                                                                                            | kernel decide        |
| <b>native_feature_normal_axis</b>   | Every literal panel normal has unit coefficient on its selected axis.                                                                                                                                                                                                                                        | kernel decide        |
| <b>solid_mesh_exact</b>             | The 2,138 literal vertices and 4,272 indexed triangles are a connected, closed oriented 6,408-edge surface; it is exactly the panel/feature construction triangle-by-triangle and has signed volume seven.                                                                                                   | <b>native_decide</b> |

| theorem                               | statement (docstring)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | method                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>boundary_sphere</b>                | Every indexed boundary edge has exactly two incident triangles, the edge-sharing triangle graph is connected, every vertex link is one cycle, and the computed counts satisfy $V - E + F = 2138 - 6408 + 4272 = 2$ . The classification theorem for connected closed surfaces therefore makes this PL surface a 2-sphere; the PL Schoenflies (Alexander) theorem then makes the bounded solid a closed 3-ball. These classical bridge theorems are imported at T3; see E. E. Moise, *Geometric Topology in Dimensions 2 and 3*, and C. P. Rourke–B. J. Sanderson, *Introduction to Piecewise-Linear Topology*. | <b>native_<br/>decide</b> |
| <b>nested_substitution_controls</b>   | The same-frame grandchild occurs at (2,2,2), and the first two nested substitution controls have 64/4,096 poses and 448/28,672 disjoint cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>native_<br/>decide</b> |
| <b>hierarchy_cell_controls</b>        | The root double refinement contains all 448 cells $4a+b$ of the seven four-by-four-by-four blocks, and every signed-permutation lower-corner action permutes the 64 offsets in the form required by equivariance.                                                                                                                                                                                                                                                                                                                                                                                              | <b>native_<br/>decide</b> |
| <b>registered_shell_cell_controls</b> | The 192 panel features map onto the 22 shell cells; every legal-contact frame is one of the 48 signed permutations; and an exact opposite center/normal/coefficient mate owns the cell beyond the root panel.                                                                                                                                                                                                                                                                                                                                                                                                  | <b>native_<br/>decide</b> |
| <b>orientation_group_24</b>           | The 19 contact frames generate exactly the literal 24-element proper cubic rotation group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>native_<br/>decide</b> |

| theorem                                           | statement (docstring)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | method                    |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>substitution_<br/>modular_<br/>coincidence</b> | The cell-labelled R44 chair substitution is a total constant-length substitution on 168 labels. Its $168 \times 168$ matrix has column sum eight and least positive power three; its least modular-coincidence depth is three. The first lexicographic witness is address $(0,0,2)$ and label 78, namely generated frame 11 and local chair cell 1. Lee-Moody, *Lattice substitution systems and model sets*, Theorem 3, supplies the external implication from primitive modular coincidence to model sets; that classification is a cited T3 import, not part of this theorem. | <b>native_<br/>decide</b> |
| <b>seed_language_<br/>closure</b>                 | Closing the legal $2 \times 2 \times 2$ block language under all 27 windows of a substituted block has successive sizes 168, 600, 1,278, 1,398 and 1,410, after which the language is stable. Exactly 27 blocks on $\{-1,0\}^3$ are fixed by their eight residue maps. They form two proper-frame orbits of sizes 24 and 3; 24 seeds have trivial origin stabilizer and three have stabilizer order 8. Every seed reproduces itself in place, and the cell substitution commutes with all 24 generated proper frames.                                                            | <b>native_<br/>decide</b> |
| <b>mesh_angle_<br/>audit</b>                      | Every actual mesh edge has one of the declared flat/carrier/feature dihedrals. The 1,536 feature edges contain 32 copies of each (base/ridge, magnitude 1..12, positive/negative side) class with the exact squared-angle and edge-length formula.                                                                                                                                                                                                                                                                                                                               | <b>native_<br/>decide</b> |

TABLE 3. The written lemmas of Sections 2–5 that formed the **Hypotheses** record, each now a Lean theorem of the pinned development (the record is empty, Appendix C): the former field, the written lemma it quoted, the theorem, its tier by its own axiom line, and the argument of the formal proof where it differs from, or refines, the written one (FIGURES.md, Table 3).

| former field                      | written lemma | Lean theorem                            | tier | how the formal proof argues                                                                                                                                                              |
|-----------------------------------|---------------|-----------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| retained_core_overlap             | A-L5.1        | retained_core_overlap_holds             | T1   | density choice of overlapping cube interiors; eighth-grid midpoint                                                                                                                       |
| carrier_feature_frame_reduction   | R§6, E5       | carrier_feature_frame_reduction_holds   | T1n  | six carrier symmetries from diameter pairs and the missing corner; signed heights from membership thresholds along normals                                                               |
| circular_cone_solid_angle         | A-L4.1        | circular_cone_solid_angle_holds         | T1   | horizontal disk slices; volume $(2\pi/3)(1 - L/\sqrt{1+L^2})$ , $L = 3/25$                                                                                                               |
| per_tube_homeomorphisms           | A§1           | per_tube_homeomorphisms_holds           | T1n  | inverse written with the continuous clamp $\psi$ ; seams and joint continuity in the height proved                                                                                       |
| feature_tube_maps_glue            | A§1           | feature_tube_maps_glue_holds            | T1n  | gluing by finite composition of supported homeomorphisms                                                                                                                                 |
| feature_tube_map_carries_carrier  | A§1           | feature_tube_map_carries_carrier_holds  | T1n  | $F(x) \in Q \Leftrightarrow x \in P$ proved pointwise, not inferred from retained cores                                                                                                  |
| feature_circular_cone_containment | A-L4.1        | feature_circular_cone_containment_holds | T1n  | agreement with the signed-tent hypograph on an open ambient neighbourhood of every feature-graph point (base edges included); Lipschitz cone inclusion transported by an affine isometry |

| former field                 | written lemma | Lean theorem                       | tier | how the formal proof argues                                                                                                                                                                                                                                                                      |
|------------------------------|---------------|------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| planar_area_carrier_recovery | R§6, E5       | planar_area_carrier_recovery_holds | T1   | alternative proof of the unchanged proposition: diameter-endpoint argument (squared distance bound $1089/100 < 12$ ; the six antipodal carrier corners recover centre and axes; the missing corner excludes reversal). Not the written planar-area calculation; say so where the paper cites R§6 |
| cone_sector_budgets          | A-L2.2        | cone_sector_budgets_holds          | T1n  | Q as a finite Boolean combination of affine halfspaces; uniform conical germs; equality with the radial tangent cone; coverage includes cone-boundary directions, disjointness is of interiors                                                                                                   |
| connected_feature_companion  | A-T4.2        | connected_feature_companion_holds  | T1n  | finite disjoint closed cover on connected sets; triple contact excluded via the discharged open-cone inclusion and volume formula (no assumption that a cone and its interior have equal volume)                                                                                                 |
| feature_containment_rigidity | A-L4.3, E1    | feature_containment_rigidity_holds | T1n  | compactness argument (K compact, $f$ an isometry, $f(K) \subseteq K \Rightarrow f(K) = K$ ) in place of the written graph-length argument; uses the distinct-tile premise                                                                                                                        |

| former field                          | written lemma | Lean theorem                                | tier | how the formal proof argues                                                                                                                                                                                                     |
|---------------------------------------|---------------|---------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>companion_pose_discrete</code>  | A-C4.4        | <code>companion_pose_discrete_holds</code>  | T1n  | recovery of centre, normal, cubic frame, eighth-grid translation; census equality via list semantics; one side condition on fixed census metadata is the named T1n theorem <code>mate_records_roles_nonempty</code> (ruling R8) |
| <code>only_registered_mates</code>    | A-L5.3        | <code>only_registered_mates_holds</code>    | T1   | endpoint keeps its frozen companion premises, including distinct placements; accepted companion/rigidity/census/overlap helpers assemble the literal 44-mate conclusion                                                         |
| <code>component_solids_cover</code>   | A-L6.2        | <code>component_solids_cover_holds</code>   | T1   | clamp to an interior point with $\text{dist}^2 \leq 3/2500 < 1/100$ , use the carried 1/4-ball and packing disjointness; alternative to assembly-wide tube gluing and conditional on an existing tiling, not an existence proof |
| <code>small_collar_realization</code> | R§8–9         | <code>small_collar_realization_holds</code> | T1n  | duplicate tube sites are identified by centre; conjugated maps agree; membership equivalence holds for every tile including unrelated ones; separation is $21/200 > 1/16$                                                       |
| <code>complete_dihedral_list</code>   | A-L3.1        | <code>complete_dihedral_list_holds</code>   | T1n  | native wedge charts and all 6,408 mesh-edge incidences; the finite incidence trace is the named R8 theorem <code>native_mesh_incidence_trace</code>                                                                             |

| former field                                          | written lemma | Lean theorem                                                     | tier | how the formal proof argues                                                                                                            |
|-------------------------------------------------------|---------------|------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>generic_<br/>feature_<br/>partner</code>        | A-L3.2        | <code>generic_<br/>feature_<br/>partner_holds</code>             | T1n  | native boundary strata and ordinary-meridian classification, with the finite exceptional-vertex tables certified by named native hooks |
| <code>baseline_<br/>component_<br/>covers_grid</code> | A-L6.1        | <code>baseline_<br/>component_<br/>covers_grid_<br/>holds</code> | T1n  | concrete companion assembly from the discharged local geometry and exact atlas                                                         |
| <code>unrestricted_<br/>alignment</code>              | A-T6.3        | <code>unrestricted_<br/>alignment_holds</code>                   | T1n  | component coverage and the certified proper-frame group assemble global registration                                                   |

TABLE 4. Axiom lines: the output of `#print axioms` for the finite theorems and the main theorems; “standard” is `propext`, `Classical.choice`, `Quot.sound`, and hooks are the per-theorem `native_decide` axioms (FIGURES.md, Table 4).

| declaration                                    | axioms ( <code>#print axioms</code> )                         |
|------------------------------------------------|---------------------------------------------------------------|
| <code>children_partition_<br/>2P</code>        | none                                                          |
| <code>contact_closure_30</code>                | hooks: <code>contact_closure_30</code>                        |
| <code>profile_canonical</code>                 | <code>propext</code> ; hooks: <code>profile_canonical</code>  |
| <code>atlas_44</code>                          | standard; hooks: <code>atlas_44</code>                        |
| <code>first_shells_33</code>                   | <code>propext</code> ; hooks: <code>first_shells_33</code>    |
| <code>central_completion</code>                | <code>propext</code> ; hooks: <code>central_completion</code> |
| <code>parent_conflicts_28</code>               | hooks: <code>parent_conflicts_28</code>                       |
| <code>parent_atlas_eq_fine</code>              | standard; hooks: <code>parent_atlas_eq_fine</code>            |
| <code>mates_census</code>                      | standard; hooks: <code>mates_census</code>                    |
| <code>deviations_distinct</code>               | none                                                          |
| <code>no_native_symmetry</code>                | <code>propext</code>                                          |
| <code>native_features_<br/>length</code>       | <code>propext</code>                                          |
| <code>native_feature_<br/>centers_nodup</code> | <code>propext</code>                                          |
| <code>native_feature_<br/>normal_axis</code>   | <code>propext</code>                                          |
| <code>solid_mesh_exact</code>                  | standard; hooks: <code>solid_mesh_exact</code>                |
| <code>boundary_sphere</code>                   | standard; hooks: <code>boundary_sphere</code>                 |

| declaration                      | axioms (#print axioms)                                                                                                             |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| nested_substitution_controls     | Quot.sound, propext; hooks: nested_substitution_controls                                                                           |
| hierarchy_cell_controls          | hooks: hierarchy_cell_controls                                                                                                     |
| registered_shell_cell_controls   | propext; hooks: registered_shell_cell_controls                                                                                     |
| orientation_group_24             | Quot.sound, propext; hooks: orientation_group_24                                                                                   |
| substitution_modular_coincidence | propext; hooks: substitution_modular_coincidence                                                                                   |
| seed_language_closure            | Quot.sound, propext; hooks: seed_language_closure                                                                                  |
| mesh_angle_audit                 | standard; hooks: mesh_angle_audit                                                                                                  |
| r44_einstein                     | standard; 21 hooks (DischargeCarrier-Strata.carrier_coordinate_states_table, DischargeMesh-Trace.native_mesh_incidence_trace, ...) |
| tiling_homochiral                | standard; 13 hooks (DischargeCarrier-Strata.carrier_coordinate_states_table, DischargeMesh-Trace.native_mesh_incidence_trace, ...) |
| tiling_chirality_corollary       | standard; 13 hooks (DischargeCarrier-Strata.carrier_coordinate_states_table, DischargeMesh-Trace.native_mesh_incidence_trace, ...) |
| carrier_hierarchy                | standard; 17 hooks (DischargeCarrier-Strata.carrier_coordinate_states_table, DischargeMesh-Trace.native_mesh_incidence_trace, ...) |
| carrier_hierarchy_unique         | standard; 17 hooks (DischargeCarrier-Strata.carrier_coordinate_states_table, DischargeMesh-Trace.native_mesh_incidence_trace, ...) |
| substitution_modular_coincidence | propext; hooks: substitution_modular_coincidence                                                                                   |

**8.2. The limit-periodic corollary.** The substitution of Section 3.1 is a lattice substitution in the sense of Lee and Moody [22]. Label each unit cell of a registered tiling by the frame of the chair that covers it and by which of the seven cells of that chair it is:  $24 \times 7 = 168$  labels. The refinement replaces each labelled cell by its eight labelled children at twice the scale, so the labelling is a substitution on  $\mathbb{Z}^3$  with inflation 2 and 168 symbols, and its  $168 \times 168$  substitution matrix has constant column sum  $8 = |\det 2I|$ .

**Theorem 8.1** (modular coincidence; T1N, replayed in Python). *The 168-label substitution is total and primitive, with least primitive exponent  $N = 3$ , and*

it has a modular coincidence of least depth  $M = 3$ : at the address  $a = (0, 0, 2)$  modulo  $2^3$ , the level-3 descendant of every one of the 168 labels has label  $i = 78$ .

The Lean theorem is `substitution_modular_coincidence`. The check tracks the full set of 168 labels from the root, so the coincidence is over all labels, not along a single orbit; the integers  $N$  and  $M$  are independent and happen to coincide.

A legal fixed point on the whole of  $\mathbb{Z}^3$  exists. A substitution-legal fixed labelling of  $\mathbb{Z}^3$  is determined by its seed on  $\{-1, 0\}^3$ : the cell  $-\delta$ ,  $\delta \in \{0, 1\}^3$ , has parent  $-\delta$  and residue  $\delta$ , so its eight labels satisfy their fixed-residue equations; conversely, a seed in the substitution language satisfying those equations yields such a labelling, whereas a block satisfying the equations but outside the language need not (one such block encodes inconsistent whole-chair labels). The substitution language contains 27 fixed seed blocks on  $\{-1, 0\}^3$ , in proper-rotation orbits of sizes 24 and 3; iterating any one gives a legal substitution-fixed labelling of  $\mathbb{Z}^3$  (Figure 12). The count is by exact closure of the language of legal  $2 \times 2 \times 2$  blocks, which grows through 168, 600, 1,278, 1,398 and 1,410 blocks under successive substitution and then is stable, and every seed is checked to reproduce itself in place under iteration (the theorem `seed_language_closure`, T1N, whose statement lists the five closure sizes, the count 27, the orbit sizes 24 and 3, the stabilizer histogram, in-place reproduction and commutation with the 24 frames; the replay step of the same name in `verify/replay.py`; and, independently, `paper/scripts/seed\_language\_closure.py`, standard library, Section A); a bounded search inside level-3 patches, which an earlier draft reported as the count 24, misses the three seeds of the small orbit, and the external review of the manuscript found them. Every finite patch of a fixed labelling occurs in some  $\sigma^n(j)$  and hence, by primitivity, inside a refinement of the native chair, that is, inside the tiling of Theorem 3.1; so the chair poses the labelling encodes form a registered chair tiling of  $\mathbb{R}^3$  all of whose contacts lie in  $\mathcal{A}_{44}$ , and small-collar realization (Section 3.4) turns it into a tiling by  $Q$  at those poses. We write  $T_w$  for the tiling determined by the seed  $w$ .

**Proposition 8.2** (symmetries of the fixed points; finite data T1N (`seed\_language\_closure`) and T2, the deduction written). *Let  $G$  be the 24-element proper cubic group, acting on the seed cube by moving the cells and composing the frames in the labels. For each of the 27 fixed seeds  $w$ ,  $\text{Sym}(T_w)$  is the stabilizer of  $w$  in  $G$ . The stabilizers have order 1 for the 24 seeds of the large orbit and order 8 for the three seeds of the small orbit; for the seed  $(28, 84, 91, 35, 98, 42, 49, 105)$ , in the repository's label order, the stabilizer is generated by  $(x, y, z) \mapsto (z, y, -x)$  and  $(x, y, z) \mapsto (-x, -y, z)$ . Hence some tilings by  $Q$  have trivial symmetry group and some have symmetry group of order 8.*

*Proof.* Let  $g = (R, t)$ ,  $x \mapsto Rx + t$ , be a symmetry of  $T_w$ . As in the proof of Theorem 7.2, registration and the absence of self-isometries give  $R \in G$  and

$t \in \mathbb{Z}^3$ . The parent partition is read from the tiling alone, and the children of a rotated pose are the rotated children (a finite check over  $G$  in the same script), so by the uniqueness of parents (Theorem 6.4) the parents of  $RT$  are the rotated parents of  $T$ ; with the translation clause of Theorem 6.5,  $D(gT) = (R, t/2) D(T)$ . The seed being fixed in place,  $D(T_w) = T_w$ , so  $(R, t/2)$  is again a symmetry of  $T_w$  and  $t/2 \in \mathbb{Z}^3$ ; iterating,  $t \in \bigcap_n 2^n \mathbb{Z}^3 = \{0\}$ . Thus  $\text{Sym}(T_w) = \{R \in G : RT_w = T_w\}$ . The substitution commutes with the action of  $G$  on labelled cells (the same finite check), so  $RT_w$  is the fixed point with seed  $Rw$ ; a fixed point is determined by its seed, so  $RT_w = T_w$  exactly when  $Rw = w$ . The orders and the generators are read off the 27 seeds.  $\square$

The seed  $(6, 0, 4, 5, 2, 3, 0, 1)$  of the large orbit is the one whose ancestors fill only an octant (Section 6); its tiling has trivial symmetry group. The three seeds of the small orbit each consist of eight whole tiles at the origin, one in each octant, each its own child 000 at every level.

We recall the definitions the corollary uses, from Lee and Moody [22]. A cut-and-project scheme consists of a lattice  $\tilde{L} \subset \mathbb{R}^3 \times G$ , where  $G$  is a locally compact abelian group, such that the physical projection  $\pi_1$  is injective on  $\tilde{L}$  and the internal projection  $\pi_2$  has dense image in  $G$ . A model set is, up to translation,  $\{\pi_1(\ell) : \ell \in \tilde{L}, \pi_2(\ell) \in W\}$ , where the window  $W \subset G$  has nonempty interior and compact closure; it is regular when  $\partial W$  has Haar measure zero. Here the internal group is  $\varprojlim_n \mathbb{Z}^3/2^n \mathbb{Z}^3$ , the 2-adic completion. A modular coincidence occurs when, at some substitution depth and residue address, all substitution maps at that address have the same output label.

**Corollary 8.3** (limit-periodic structure of the fixed point; the cited imports are T3). *By Lee and Moody’s Theorem 3 [22], applied in the direction (iv)  $\Rightarrow$  (ii) with its hypotheses verified as above (Theorem 8.1: primitivity; Perron–Frobenius eigenvalue  $8 = |\det 2I|$ ; the label classes of the fixed point partition  $\mathbb{Z}^3$ ), the 168 label classes of the substitution fixed point are regular model sets for the 2-adic cut-and-project scheme of [22]; by Schlottmann’s theorem, in the form stated by Lee, Moody and Solomyak [24, Theorem 5.11], each is pure point diffractive; in the terminology of Baake, Moody and Schlottmann and of Baake and Grimm [3–5] the structure is limit-periodic.*

The corollary is a statement about the fixed point, not about every tiling by  $Q$ : whether every tiling lies in the substitution hull is the fault-line question left open in Section 6, and Lee and Moody’s own passage from the substitution to every tiling obeying the Taylor–Socolar matching rules is a separate three-part argument [23], not an application of their Theorem 3. In Socolar’s words about the hat, whose structure “is qualitatively different from the limit-periodic structure of the hexagonal Taylor–Socolar monotile tiling, which has Bragg diffraction peaks at wavelengths of the form  $2^{-n}k_0$  for arbitrarily large  $n$ ” [38], the fixed-point label-class structure established here is limit-periodic with a 2-adic internal space, rather than quasicrystalline in the sense contrasted by Socolar. Akiyama and Lee verified pure point spectrum

for a 168-prototile Taylor–Socolar substitution by an overlap coincidence [2], and Lee and Moody checked the sphinx’s modular coincidence by computer at depth 8 [22]; ours is checked at depth 3 in Lean and replayed in Python.

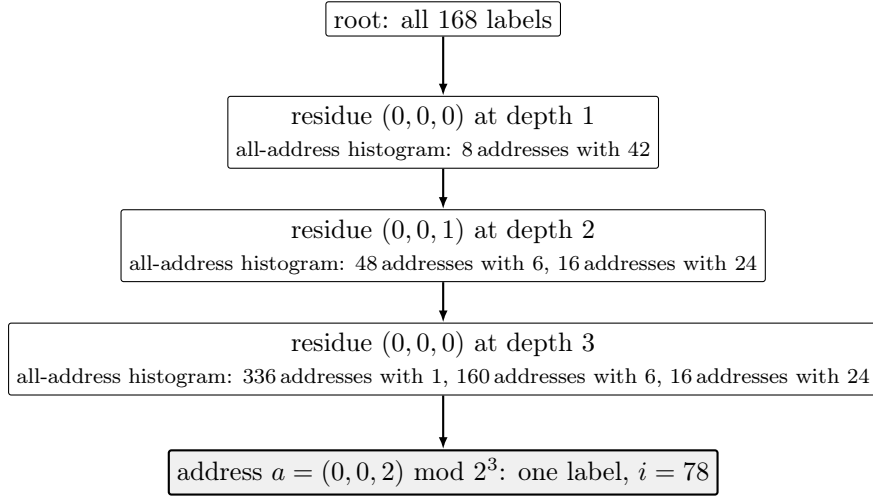


FIGURE 12. The modular-coincidence address and the all-address distributions of label-set sizes at depths 1–3.

## 9. FINITE ADMISSIBILITY AND CONTEXTUAL DISTINCTIONS

The results above give a concrete instance of a distinction in the Six Birds (SBT) calculus [43]: a rule can survive composition while its realizations require successively finer contextual distinctions. This complementary reading proceeds from a local obstruction to the intrinsic hierarchy, then asks what the construction suggests about descriptions of collective systems. The deductions below are written proofs; no new Lean verification is claimed.

*A distinction that cannot be erased.* Let  $C(x, y, z) = (y, z, x)$ . Since  $CP = P$ , the nine pairs  $(C^a Q, C^b Q + (1, 1, 1))$ , for  $a, b \in \{0, 1, 2\}$ , have the same ordered pair of bare carriers. Their feature profiles match precisely when  $a = b$ . The exact checker `paper/scripts/contextual_frames.py` reconstructs the 24 opposed feature centres in each pair from the solid's rational feature records: every off-diagonal pair mismatches all 24 coefficients and has an interior overlap. For example,  $Q$  and  $CQ + (1, 1, 1)$  both contain  $(5/4, 11/8, 1)$  in their interiors: there the root's role 188 protrudes in  $+z$  by  $12/10000$ , and the neighbour's role 4 protrudes in  $-z$  by  $5/10000$ . The diagonal pairs are rotated copies of the 000/central subdivision pair. In SBT terminology this is a *descent obstruction*: admissibility is not a function of the erased carrier description. Testing the neighbour against each of the three root frames requires retaining all three neighbour-frame distinctions.

*Two different closure questions.* SBT's *sufficiency closure* (F7 of [43]) asks whether each declared readout factors through an observation. It does not assert that an operation preserves legal configurations. Here the latter is the separate geometric statement  $D(X_Q) \subseteq X_Q$ , where  $X_Q$  is the space of tilings by  $Q$  (Theorem 6.5). This permits repeated recognition and restandardization under the same rule. The observations of position in that hierarchy behave differently.

**Proposition 9.1** (phase distinctions). *Fix a registered tiling  $T$  in integer coordinates. Its level- $n$  parent anchors lie in a unique coset  $\alpha_n(T) + 2^n \mathbb{Z}^3$ , with  $\alpha_{n+1}(T) \equiv \alpha_n(T) \pmod{2^n}$ . The observations*

$$q_n^T(v) = v - \alpha_n(T) \pmod{2^n}, \quad v \in \mathbb{Z}^3,$$

*have  $8^n$  values. Writing  $\ker q = \{(v, w) : q(v) = q(w)\}$ ,*

$$\ker q_n^T = \{(v, w) : w - v \in 2^n \mathbb{Z}^3\}, \quad \ker q_{n+1}^T \subsetneq \ker q_n^T, \quad \bigcap_{n \geq 0} \ker q_n^T = \Delta,$$

*where  $\Delta = \{(v, v) : v \in \mathbb{Z}^3\}$ .*

*Proof.* The common-parent parity argument of Theorem 6.5, applied at every level and expressed in the original coordinates, gives the cosets. Each parent has child 000 at its own anchor, so the anchor sets are nested and the cosets are coherent. Reduction modulo  $2^n$  is onto and has the displayed kernel. The pair  $v, v + 2^n e_1$  lies in that kernel but not the next; their intersection is  $\Delta$  because  $\bigcap_n 2^n \mathbb{Z}^3 = \{0\}$ .  $\square$

The phases are read from the intrinsic hierarchy, locally at each fixed level with a neighbourhood growing with the level. In particular, for  $p \in \mathbb{Z}^3$ ,  $\alpha_n(T + p) = \alpha_n(T) + p \pmod{2^n}$ . A period must therefore vanish modulo every  $2^n$ , recovering Theorem 7.1. This uses the global partitions, not exhaustion by one tile's ancestors. Equal phases need not give equal patches:  $8^n$  counts phase states, not patch types or entropy. Each next level requires eight times as many phase states, while each fixed phase observation has the lawful translation update  $q_n^T(v + u) = q_n^T(v) + u$ .

*Exact contextual sufficiency.* Let  $\ell : \mathbb{Z}^3 \rightarrow A$  be the 168-label cell encoding of  $T$  (Section 8.2). It is faithful: a cell's position, native frame and native cell identify its owner's whole pose. Declare the readouts to be all translated labels,  $v \mapsto \ell(v + u)$  for  $u \in \mathbb{Z}^3$ . Their common equivalence, the *predictive quotient* for this spatial probe, satisfies

$$v \equiv w \iff \forall u \in \mathbb{Z}^3, \ell(v + u) = \ell(w + u) \iff w - v \in \text{Per}(\ell).$$

Indeed, put  $u = z - v$ ; the reverse implication is the definition of a period. Faithfulness gives  $\text{Per}(\ell) = \text{Per}(T) = \{0\}$ . Thus any observation sufficient for all these readouts must distinguish every probe origin. This consequence holds for any faithful aperiodic lattice labeling; Chair44 additionally supplies the intrinsic phase hierarchy organizing such distinctions.

This sufficiency requirement concerns the declared probes. It excludes neither finite descriptions nor useful coarse models. The coherent phases give a 2-adic coordinate, but need not determine the whole tiling; the hull and diffraction questions retain the scope stated in Section 8.2.

*Collective descriptions and their limits.* These results suggest a broader question: which relationships must a description retain for a collective to support its declared operations? An inventory of parts can identify arrangements that behave differently when placed in a further context. The appropriate criterion is therefore relative to a specified family of contexts  $\mathcal{C}$  and their observable outcomes:

$$q(x) = q(y) \implies \text{Outcome}(C[x]) = \text{Outcome}(C[y]) \quad \text{for every } C \in \mathcal{C}.$$

Here the outcome includes whether the composition is admissible at all. The coarsest equivalence satisfying this criterion identifies exactly those states that all declared contexts fail to distinguish. It specifies the relational information required by those tests, rather than a complete microscopic inventory.

In SBT terms, a separating context witnesses a descent obstruction. Enlarging the context family can only refine its observational equivalence, but strict refinement at arbitrarily large depths requires a separate argument. Recursive composition alone does not supply one. For Chair44, Theorem 9.1 supplies that argument: each phase level identifies positions separated by some nonzero translations, while no such identification survives every level. Independently, coarsening preserves the geometric admissibility law that makes the hierarchy repeatable.

The philosophical consequence is precise and limited: a finite, recursively stable law can require a realization to preserve relational distinctions at arbitrarily large scales. The infinite arrangement carries those distinctions; the law constrains their organization. This suggests a question for other collective systems: which permitted interaction reveals information erased by a proposed description, and does retaining that information suffice after further composition? An answer requires identifying that system's own states, operations and separating contexts. Chair44 provides a geometric instance in which both lawful recursion and unbounded contextual differentiation can be established.

## 10. REMARKS AND OPEN PROBLEMS

We close with questions, in the hat paper's manner, and we phrase each as a question.

- (1) *Symmetry.* The fixed-point construction (Proposition 8.2) yields tilings with symmetry groups of orders 1 and 8. Is the bound  $|\text{Sym}(T)| \leq 24$  attained, and which other finite symmetry groups occur? Question 32 of Coulbois, Gajardo, Guillon and Lutfalla asks for a geometric monotile whose tilings all have trivial stabilizer [10];  $Q$  is not one (Section 7).
- (2) *Fault surfaces and the tiling space.* The ancestors of one fixed tile can fail to cover space (Section 6). Which non-exhausting hierarchies occur, and does every tiling lie in the substitution hull? The hat paper asks the latter of hats [36]; for  $Q$  the question is also where a hull-level version of Corollary 8.3 (Section 8.2) would have to be settled.
- (3) *Convexity.* Is there a convex strongly aperiodic monotile in  $\mathbb{R}^3$ ? With reflections forbidden, the biprism is convex and weakly but not strongly aperiodic; with reflections allowed it admits periodic tilings;  $Q$  is strongly aperiodic and not convex; and for translations alone convex tiles cannot be aperiodic [10].
- (4) *Achirality.* Is there a strongly aperiodic monotile of  $\mathbb{R}^3$  with a mirror symmetry, or one whose tilings mix both handednesses? The Spectre paper poses the einstein problem for each isometry group  $G$  [37];  $Q$  answers it for the full isometry group of  $\mathbb{R}^3$  and, being strictly chiral, for the group of proper motions at the same time.
- (5) *Simplicity.* What is the smallest number of faces, or of vertices, of a strongly aperiodic monotile in  $\mathbb{R}^3$ ?  $Q$  has 4,272 triangles because each of its 192 features is a pyramid; within the twelve-parameter family of Section E.6 the proof does not change, and we do not know how far the architecture can be simplified.
- (6) *Heesch and isohedral numbers; decidability.* What Heesch numbers and isohedral numbers occur for polyhedra in  $\mathbb{R}^3$ , and is it decidable whether a given polyhedron tiles  $\mathbb{R}^3$ ? We prove nothing in this direction.
- (7) *A shorter proof.* Simplified proofs of the hat's aperiodicity appeared within months [1]. The universality half of our proof, Sections 4 and 5, is

the part we would most like to see shortened; the finite half is a census that a different argument might avoid.

- (8) *The mechanism elsewhere.* The construction puts small features on a periodic carrier so that the features force a registered lattice substitution whose parent language is its own fine language. Does the same design yield einsteins in other dimensions, on other lattices, or in other crystallographic settings, and is there a version in the plane with a different carrier?

The features of  $Q$  have height unit  $1/10000$  and maximum height  $12/10000$ , measured in carrier-cell units. Everything the theorem says about tilings by  $Q$ , that they exist, that they are hierarchical, that they have no period and no symmetry of infinite order, and that they never mix a tile with its mirror image, is decided by those pyramids. We do not know whether fewer or larger features, or a fundamentally simpler solid, can enforce the same conclusions;  $Q$  may be far from the simplest example.

#### ACKNOWLEDGEMENTS

The repository archives two external adversarial review rounds of the development and one of this manuscript; the latter corrected the count of fixed seeds and supplied the first non-exhausting and the first symmetric fixed points (Sections 6 and 8.2), and its report is filed with the paper’s review record. Automated reasoning agents also checked the development, certificates and manuscript under the recorded review protocol. The remaining errors are ours. We thank the authors of the hat and Spectre papers for the form of the definitions, of the theorems and of the disclosure that this paper follows, and the authors of Lean 4 and Mathlib.

#### USE OF AI SYSTEMS

This paper and the work it reports were produced with artificial-intelligence systems under the author’s direction; the author takes responsibility for every statement, and no AI system is an author. *The discovery.* The solid was found by an OpenAI reasoning model (ChatGPT, Astra) that was given the Six Birds framework, the author’s emergence calculus [43–45], together with the construction record of the repository, and that worked from the reading of the aperiodic-monotile phenomenon described in Sections 1.3 and 3.3: the framework was the input, the solid the output. Its six hash-chained packets (Section 3.3, *provenance*/) contain the periodic control, the frame change, the 44-contact atlas and the written alignment proof; the construction record that preceded them was likewise produced by automated agents under the author’s direction. *The verification.* The Lean formalization, the finite certificates and the negative controls were implemented by automated coding agents (OpenAI Codex) and reviewed by agents of a different model family (xAI Grok; OpenAI Codex on a second model) under the protocols recorded in the repository, and the two external adversarial review rounds of Section 8.1 and the external review of this manuscript were carried out by such agents. *The manuscript.*

This text was largely written by Claude Fable 5.1 (Anthropic) from the author’s plan, the repository’s proofs and certificates and the literature corpus, and was reviewed step by step by an independent automated reviewer (OpenAI Codex, gpt-6-astra) whose verdicts are archived with the review package; the author directed, read and edited every part.

## APPENDIX A. DATA AND REPLAY

*Inputs.* The theorem is about one solid, `solid/r44_solid.json`: its 2,138 vertices as exact rationals, its 4,272 triangles, the 192 features with their signed coefficients, and the eight child poses. Two certificate files carry the finite data the proof uses: `certificates/candidate_certificate.json` (the contact closure and profile, the 44-contact atlas, the 33 first shells, the completions and conflicts, the parent census) and `certificates/companion_collision_certificate.json` (the 5,273 rejection records with their 299,975 collision witnesses, the boxes themselves in a companion file). Their sha256 digests are in Table 5. The replay entrypoint checks that the four canonical inputs are byte-identical to the packet copies and checks the fixed solid digest before invoking the checkers.

TABLE 5. Input hashes and tool versions, computed at the state of the repository described in this paper (FIGURES.md, Table 6).

| artifact                                                       | sha256 / version                                       |
|----------------------------------------------------------------|--------------------------------------------------------|
| <code>solid/r44_solid.json</code>                              | f320d7a0c2d784a3...42f0ed55                            |
| <code>solid/native_panels.csv</code>                           | b9eec9fd834abba4...d3a9ed2d                            |
| <code>certificates/candidate_certificate.json</code>           | 44e9b3f6048497de...c5de5ff2                            |
| <code>certificates/companion_collision_certificate.json</code> | 36e89e2fe81f4788...1b068b08                            |
| <code>certificates/collision_core_witnesses.jsonl.gz</code>    | ae64a5283400c61e...161f8a45                            |
| Lean toolchain                                                 | leanprover/lean4:v4.31.0                               |
| Mathlib commit                                                 | fabf563a7c95                                           |
| batteries / aesop / Qq / proofwidgets                          | fa08db58b30e, e3cb2f741431, f46324995fca, 24b0d9dc081c |
| <code>verify/replay.py</code> (Python standard library)        | PASS, 27.0 s                                           |
| Python (replay and tables)                                     | 3.12.3                                                 |

*Replay in one minute.* From the repository root,  

```
python3 verify/replay.py
```

runs, with the Python standard library only and no network, the checker of the registered theorem (the first implementation), the alignment checker

and the integer-only replay of the collision boxes (the second implementation), the mesh check `verify/boundary_sphere.py`, the tube-formula controls, `verify/substitution_modular_coincidence.py` and the block-language closure `verify/seed_language_closure.py`; it asserts the input digests first and writes `verify/replay_report.json`. In the recorded clean-clone run at the pinned commit (`paper/review/REPLAY_FROM_PAPER.md`, 2026-09-10) the replay completed in about half a minute (28.316 seconds wall time) on the machine that ran the Lean builds; its hardware is not otherwise specified; hardware specifications and Lean build time and peak memory are not reported for this version. Three further commands complete the README’s one-minute list:

```
(cd lean/R44 && python3 gen/extract.py --check)
python3 -m unittest discover simulations/tests
cd verify/packets/r44_unrestricted_alignment
python3 src/test_mutations.py
```

The first checks that the Lean literals generated from the JSON inputs match those inputs. The unit-test command reported 10 tests with six skipped by optional-dependency guards; it does not establish that those six tests passed. The last command rejects six corrupted inputs (a missing triangle, a reversed triangle, a moved apex, a missing rejection witness, a wrong companion quantifier, a missing registered contact) and states its own scope: it checks that malformed inputs are rejected and does not validate the continuous proof.

*Paper-lane scripts.* The standard-library scripts cited in the text as T2 live in `paper/scripts/`: the exact closure of the block language and the fixed seeds (`seed_language_closure.py`, Section 8.2), the planar areas of Section E.3 (`planar_areas.py`) and the non-convexity witness; each reads the repository data and prints `STATUS: PASS`.

*Figures.* Figures 1 and 4 are drawn by `figures/print-preview/build\_preview.py` from `solid/r44\_solid.json` with the exaggerations stated in their captions: the carrier and the 192 feature centres keep their coordinates, every feature base is widened  $\times 5$  about its centre and every signed height is multiplied by 80, so polarities and relative heights are preserved and nothing is added, omitted or moved. The script checks all 192 feature records against the mesh and the native panel table in rational arithmetic, and a second script reads the display mesh back and reconstructs its 768 feature triangles independently; the outputs and their digests are in `figures/print-preview/`. Figures 5 and 6 are the renders of `figures/render\_r44.py`. No figure is verification.

*Exact arithmetic.* Every certificate is integer or rational data; the only decimal literals in the certificates are two timing fields. The checkers use Python integers and `fractions.Fraction`; the only floating-point conversion in the repository’s verification code is the export of the mesh to a rendering format. The verification scripts named in the “Paper-lane scripts” paragraph

also use exact rationals only; the planar-area checker bounds each non-axis facet's area  $\sqrt{q}/2$  by  $(q + m^2)/(4m)$  with an integer-square-root witness  $m$ , exactly. Collision boxes are stored in integer units of  $1/400$ . The Lean theorems decide integer and rational literals.

*Lean.* The development is `lean/R44`, built with `lake build` under the toolchain of Table 5 (the build is capped at four parallel jobs and the Mathlib cache is pre-seeded); `lean/R44/scripts/controls.sh` runs the positive build, then every must-fail file against its expected diagnostic, the scope regressions and the admission count, in one command. The axiom lines of Table 4 are produced by the build itself (`build_axioms.log`).

*Viewer.* `viewer/index.html` shows the solid, the eight-child cluster and the 64-, 512- and 4,096-copy patches in the browser without a server. It is generated from the canonical rational data with the source hashes recorded; its features are drawn exaggerated by default, with a switch to true heights; colours are annotation. It also offers a period test on finite patches, which, as its own caption says, proves nothing.

*Code.* Repository: <https://github.com/ioannist/six-birds-tiles>. The Lean statements, hypothesis record and axiom lists reproduced in this paper refer to commit `d90313a717`, and the recorded clean-clone replay was run at that commit (an earlier recorded run at `bdc6d3b957`, before the block-language closure joined the replay, gave the same digests and outcomes). The intended paper-release tag is `paper-v1`; tag creation and licensing remain pending owner actions.

## APPENDIX B. CENSUS TABLES

Table 6 collects the finite censuses used in the proof, read by a script from the certificates and the packet results that the replay checks. For each row the Lean theorem that certifies it is: the contact closure and profile, `contact_closure_30` and `profile_canonical`; the atlas and the group it generates, `atlas_44` and `orientation_group_24`; the first shells, completions and conflicts, `first_shells_33`, `central_completion` and `parent_conflicts_28`; the parent census, `parent_atlas_eq_fine`; the companion census and its collisions, `mates_census`; the substitution, `substitution_modular_coincidence`. The companion census is computed by the two independent implementations of Section 1.4; the second implementation records, for its own audit, that it imports no upstream program, derives the features from the actual triangles, uses no flat-panel filter to reject placements, and uses no existing rejection to prune the companion options.

The 2,388 candidate contact poses arise as follows. A root chair in the identity pose has 22 grid cells adjacent to it across its 24 panels (the 21 outer panels face distinct exterior cells, while all three notch panels face the missing cell  $(1, 1, 1)$ , giving 22 distinct shell cells); a neighbouring chair in one of the 48 signed frames covering a shell cell is positioned by subtracting each of its seven cells from that shell cell; deduplicating and rejecting overlaps

TABLE 6. The censuses (FIGURES.md, Table 2).

| census                                                                                                          | count                                       |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| contact closure: internal face contacts $\rightarrow$ closed states; sign equations; balanced components        | 21 $\rightarrow$ 30; 372; 12                |
| shell poses in 48 frames $\rightarrow$ legal contacts (the atlas); distinct rotations; group generated          | 2,388 $\rightarrow$ 44; 19; 24              |
| first shells; outer-root completions found / impossible; parent conflicts                                       | 33; 14/18; 28                               |
| parent contacts: candidates $\rightarrow$ disjoint $\rightarrow$ legal; halved = fine atlas                     | 697 $\rightarrow$ 116 $\rightarrow$ 44; yes |
| complete-feature mate poses $\rightarrow$ isolated $\rightarrow$ survivors                                      | 6,862 $\rightarrow$ 5,317 $\rightarrow$ 44  |
| rejected isolated poses; companion-option collisions checked                                                    | 5,273; 299,975                              |
| substitution labels; column sum; least primitive exponent $N$ ; coincidence depth $M$ , address $a$ , label $i$ | 168; 8; 3; 3, (0, 0, 2), 78                 |

with the root leaves 2,388. The 6,862 companion poses are the formula poses of Corollary 4.8 over all ordered pairs of opposite features and all signed permutations with matching normals; of them 1,545 collide with the root directly, 5,317 are isolated (5,234 with a fractional translation, 83 integral), 5,273 are rejected by the collision witnesses, and the 44 survivors equal the atlas as a set.

#### APPENDIX C. LEAN STATEMENT LISTING AND AXIOMS

*The main theorem.* In `lean/R44/R44/LogicalSpine.lean`, with  $Q$  the solid defined from the canonical panel data, `Tiling Q` the tilings of  $\mathbb{R}^3$  by affine-isometric copies of  $Q$  (reflections allowed, local finiteness not assumed), and `Per`, `Sym` as in Section 1.5, the main theorem reads

```
theorem r44_einstein : Nonempty (Tiling Q)  $\wedge$   $\forall$  T : Tiling Q,
  Per T = {0}  $\wedge$  (Set.univ : Set (Sym T)).encard  $\leq$  24
```

and takes no hypothesis argument. The companion declarations of Section 8.1 (homochirality, the chirality corollary, the hierarchy and its uniqueness) retain an argument `H` of type `Hypotheses`, but that record is empty and is instantiated by the empty anonymous constructor; it imposes no mathematical assumption. The finite theorems of Table 2 take no hypothesis argument.

*The record.* The development was written around a record `Hypotheses` whose fields quoted the written lemmas of this paper, one field per lemma, with `r44_einstein` proved over that record. Each field has since been proved as a theorem of the development (Table 3; the theorems live in `lean/R44/R44/Proved/`), and at the pinned commit the record has no fields: its declaration is, verbatim,

```
structure Hypotheses : Prop where
```

The conditional theorem survives, over the empty record, as `r44_einstein_of_hypotheses`; the main theorem is its instance at the empty record. `lean/R44/HYPOTHESES.md` gives, for each former field, the written lemma it quoted, the theorem that discharged it and the delivery it came in. The diagonal audit, made after the first external review, distinguishes assertions about distinct companion tiles from relations with legitimate reflexive cases.

*Axioms.* Table 4 is the output of `#print axioms`; the main theorem’s line is, verbatim,

```
'R44.r44_einstein' depends on axioms: [propext, Classical.choice, Quot.sound,
  R44.atlas_44._native.native_decide.ax_1_1,
  R44.central_completion._native.native_decide.ax_1_1,
  R44.contact_closure_30._native.native_decide.ax_1_1,
  R44.first_shells_33._native.native_decide.ax_1_1,
  R44.hierarchy_cell_controls._native.native_decide.ax_1_1,
  R44.mate_records_roles_nonempty._native.native_decide.ax_1_1,
  R44.mates_census._native.native_decide.ax_1_1,
  R44.mesh_angle_audit._native.native_decide.ax_1_1,
  R44.nested_substitution_controls._native.native_decide.ax_1_1,
  R44.orientation_group_24._native.native_decide.ax_1_1,
  R44.parent_atlas_eq_fine._native.native_decide.ax_1_1,
  R44.parent_conflicts_28._native.native_decide.ax_1_1,
  R44.profile_canonical._native.native_decide.ax_1_1,
  R44.registered_shell_cell_controls._native.native_decide.ax_1_1,
  R44.solid_mesh_exact._native.native_decide.ax_1_1,
  R44.transported_role_geometry._native.native_decide.ax_1_1,
  R44.DischargeCarrierStrata.carrier_coordinate_states_table
    ._native.native_decide.ax_1_1,
  R44.DischargeMeshTrace.native_mesh_incidence_trace
    ._native.native_decide.ax_1_1,
  R44.DischargeVertexData.carrier_vertex_lookup._native.native_decide.ax_1_1,
  R44.DischargeVertexData.feature_index_bounds._native.native_decide.ax_1_1,
  R44.DischargeVertexData.feature_vertex_lookup._native.native_decide.ax_1_1]
```

(one line in the log, broken here, the two longest names at their namespace dot). Each `native_decide` hook trusts the compiled evaluation of one closed decidable proposition. Of the 21 hooks, 15 belong to the finite theorems of Table 2 and to the auxiliary `transported_role_geometry` check; the other six are closed checks over the certified finite data made by the discharge proofs: a side condition on the census metadata (`mate_records_roles_nonempty`), the 6,408-edge mesh incidence trace (`native_mesh_incidence_trace`), a 125-state classifier of carrier coordinates (`carrier_coordinate_states_table`), and three lookups in the tables of exceptional feature and carrier vertices. None of them changes the proposition it certifies; the repository’s rulings R8 and R10 in `lean/R44/MECHANIZATION\_PLAN.md` record why each is a named hook rather than a kernel `rf1`. The independent Python replays check the certificates and geometric data described in Section A. No `sorryAx` occurs anywhere in the development, and the directory `lean/R44/R44/Discharge/`, which held the separately built discharge library with its marked admissions at the earlier baseline, holds no module at the pinned commit.

*Controls.* `lean/R44/scripts/controls.sh` runs the positive build and then compiles every must-fail file, checking that each fails with its expected

diagnostic. The negative controls exercise the repaired diagonal application, corrupted finite inputs, mixed-handedness claims, a second-hierarchy claim and a claim that a raw geometric hierarchy has a noncanonical level-one pose. The runner also checks the scope regressions (which include legitimate reflexive uses) and reports the admission count, zero at the pinned commit. A transcript of the controls run, with build time and peak memory, is not included at the state described here.

#### APPENDIX D. THE PLANAR CALIBRATION

The same pipeline was run on the hat as a calibration of the tooling, not as a result: the hat polykite with its thirteen vertices was reconstructed as an exact polygon, its published substitution and recognition grammar were replayed with explicit proof dependencies, the growth of patches was derived, and controls were run. The calibration checked an abstract theorem that transports a period unchanged through coarsening levels and excludes nonzero periods using unbounded lower bounds on their size; its geometric and recognition inputs remain explicit imports from the hat paper [36], to whose authors the hat and its aperiodicity are due. The R44 proof uses the related obstruction in a different normalization: periods halve while nonzero registered periods have a uniform positive lower bound. The calibration is recorded in the repository under `calibration_2d/` with its own charter, receipts and lessons, and it served as the template for the Lean skeleton of Section 8 and as an exercise of the replay, control and review discipline of Section A on a known einstein.

#### APPENDIX E. GEOMETRIC DETAILS OF THE CONSTRUCTION

The recipe in Section 2 specifies the solid. Here we record how its signed height table is obtained and why the resulting boundary, asymmetry and feature angles have the properties used by the proof. The numerical choices in the recipe are one explicit member of the family described at the end of this appendix.

**E.1. How the height table is determined.** The table of heights is not arbitrary; it is determined by the substitution of Section 3.1 and can be reconstructed from it. Start with the 21 face contacts that occur inside the eight-child dissection of the doubled chair and propagate them through refinement until the set of contacts is closed; it closes at 30. At every pair of marks that these contacts bring into coincidence impose that the two heights be opposite,  $a_u = -a_v$ . This gives 372 distinct equations whose signed graph on the 192 marks has exactly twelve balanced connected components, each with eight positive and eight negative vertices. Assign the magnitude  $j$  to the  $j$ -th component in order of its smallest mark, take that mark positive, and transport signs along the graph. The result is the table of 192 entries in the solid file, and the twelve magnitudes  $1, \dots, 12$  are the twelve components.

Because every component uses a distinct magnitude, two features fit face to face only if they have the same magnitude and opposite sign.

**E.2. Boundary and topology.** The boundary of  $Q$  is a triangulated surface with 2,138 vertices, 6,408 edges and 4,272 triangles (Table 7), checked triangle by triangle against the construction in Section 2. Every edge lies in exactly two triangles, the triangle graph is connected, the link of every vertex is a single cycle, and  $V - E + F = 2$ ; these are finite facts, verified in Lean and independently by a script. Two classical results then apply, and we state them exactly as used: a connected closed triangulated surface with Euler characteristic 2 is homeomorphic to the sphere, since an orientable surface of genus  $g$  has  $\chi = 2 - 2g$  and a non-orientable one with  $k$  crosscaps has  $\chi = 2 - k$ ; and, by the polyhedral Schoenflies theorem of Alexander, a polyhedral 2-sphere in  $\mathbb{R}^3$  bounds a piecewise-linear 3-ball [26, 34]. Hence  $Q$ , the closure of the bounded domain its boundary mesh bounds, is a closed topological 3-ball. The same conclusion follows from the tube construction in Section E.5, which exhibits an ambient homeomorphism carrying  $P$  to  $Q$ .

TABLE 7. Mesh facts of  $Q$ , computed from the solid file with exact arithmetic and cross-checked against its recorded counts; the verified statements are `boundary_sphere`, `mesh_angle_audit` and `verify/boundary_sphere.py` (FIGURES.md, Table 5).

| quantity                                                                  | value                 |
|---------------------------------------------------------------------------|-----------------------|
| vertices $V$                                                              | 2,138                 |
| edges $E$                                                                 | 6,408                 |
| triangles $F$                                                             | 4,272                 |
| $V - E + F$                                                               | 2                     |
| volume (exact, divergence formula)                                        | 7                     |
| features (square pyramids)                                                | 192                   |
| feature edges; classes (magnitude, base/ridge, sign)<br>× edges per class | 1,536; $48 \times 32$ |
| base half-width $\eta$ ; height unit                                      | 1/100; 1/10000        |

**E.3. The solid has no nonidentity self-isometry.**  $Q$  has no nonidentity self-isometry. The argument has three parts, and we say once here at what level each is verified. First, an isometry  $g$  with  $gQ = Q$  preserves the carrier: the nine large planar boundary regions of  $Q$ , one for each coordinate plane at heights 0, 1 and 2, carry planar areas 2492/625, 623/625 and 1869/625 respectively (the area-four, notch area-one and area-three regions less their feature bases; exact values from the mesh, `paper/scripts/planar_areas.py`), each more than 9/10, whereas all the non-axis pyramid facets together have area less than  $1212/15625 < 1/4$ . So  $g$  permutes those nine planes; the

three distinct area types identify the height-0, height-1 and height-2 plane triples, hence  $g$  fixes the origin, the corner  $(2, 2, 2)$  and the notch, and maps  $P$  to itself (the Lean theorem `planar_area_carrier_recovery_holds`, T1, proves the same statement by a different route: a point in a feature tube is at squared distance at most  $1089/100 < 12$  from every point of  $Q$ , so the diameter pairs of  $Q$  lie in the undeformed carrier and are its antipodal corner pairs, which  $g$  permutes; the six corners recover the centre and the axes, and the preserved missing corner excludes a reversal; the planar-area calculation above is the written proof). Second, an isometry that preserves both  $Q$  and the chair has a signed permutation matrix as its linear part (the chair's own symmetry group is the six coordinate permutations) and must carry the table of feature centres and heights to itself; this reduction to one of the 48 signed frames is a Lean theorem using `native_decide`. Third, among the 48 signed frames only the identity preserves the table of 192 features (a Lean theorem by kernel `decide`). Consequently the stabilizer of a tiling's family of placements equals its symmetry group, as noted in Section 1.5.

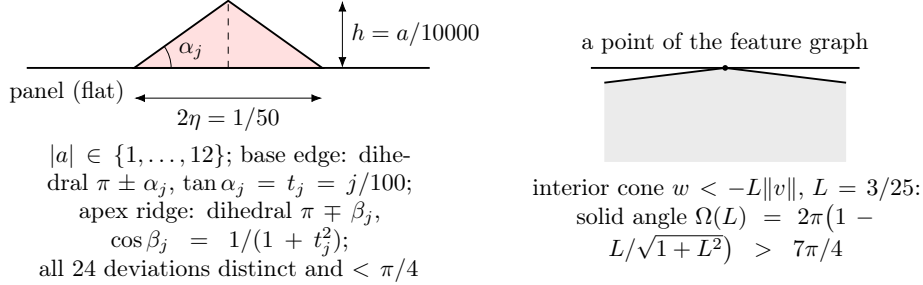


FIGURE 13. Schematic feature geometry, with exaggerated height: a pyramid of height  $j/10000$ , base half-width  $1/100$ , its deviation angles, and the interior cone with slope bound  $L = 3/25$  used in Section 4.

**E.4. Feature angles.** The 192 features play 192 distinct roles: their centres are distinct points of  $\frac{1}{8}\mathbb{Z}^3$ , and the twelve magnitudes give two sequences of twelve deviation angles, at the base edges and at the apex edges of a pyramid of slope  $t_j = j/100$ , all 24 distinct: a coincidence between the two sequences would force  $1 + (i/100)^2 = (1 + (j/100)^2)^2$ , that is  $10000 i^2 = 20000 j^2 + j^4$ , which has no solution with  $1 \leq i, j \leq 12$  (a Lean theorem by kernel `decide`). Every one of the 6,408 mesh edges carries a declared dihedral angle, flat or orthogonal except at the features; the 1,536 feature edges fall into 48 classes of 32, one class per deviation and sign (Table 7).

**E.5. Local deformation and retained cores.** Around each feature take the normal tube of half-height  $\rho = 1/100$  over its base. The map  $(u, z) \mapsto (u, z + \chi(z)f(u))$ , where  $f$  is the signed tent function of the pyramid and  $\chi(z) = \max(0, 1 - |z|/\rho)$ , is strictly increasing in  $z$  because  $\|f\|_\infty \leq 12/10000 < \rho$ ,

equals the identity on the boundary of the tube, and carries the flat base to the pyramid. The 192 tubes are pairwise disjoint (the marks on a panel are at least  $1/8$  apart in one tangent coordinate and more than  $1/5$  from every panel edge, while the bases are  $1/50$  across), so the maps glue to a homeomorphism of  $\mathbb{R}^3$  that is the identity outside the tubes and carries  $P$  onto  $Q$ . In particular every unit cube of the chair keeps its core  $[\rho, 1 - \rho]^3$  inside  $Q$ , a fact used in Section 5.

**E.6. The open parameter family.** Finally, the integer heights are a convenient member of a family. Let the twelve component heights be real numbers  $c_1, \dots, c_{12}$ , keep the signs of the graph, and use height  $c_j/10000$ . With  $t_j = |c_j|/100$ , the whole proof goes through unchanged provided every  $c_j$  is nonzero and the magnitudes  $|c_j|$  are distinct;  $1 + t_i^2 \neq (1 + t_j^2)^2$  for all  $i, j$ ; every  $t_j < 1$ , every  $(1 + t_j^2)^2 < 2$ , and  $63 \max_j t_j^2 < 1$ ; and the heights stay below  $\rho$  and  $\eta$ . These are finitely many strict inequalities satisfied by  $c_j = j$  (Table 8), so Theorem 1.2 holds on an open twelve-parameter family of solids with the same architecture. This is not stability under arbitrary perturbation of the boundary, and we claim none.

TABLE 8. The parameter family: the exact open conditions of R§11 and their values at the explicit member  $c_j = j$ , checked with exact rationals (FIGURES.md, Table 7).

| condition (R§11)                                                     | at $c_j = j$                                                                |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------|
| (1) every $c_j \neq 0$ , magnitudes $ c_j $ distinct                 | $ c_j  = 1, \dots, 12$ , distinct                                           |
| (2) $1 + t_i^2 \neq (1 + t_j^2)^2$ for all $i, j$                    | holds; smallest gap $39/390625$                                             |
| ( $t_j =  c_j /100$ )                                                |                                                                             |
| (3) $t_j < 1$ ; $(1 + t_j^2)^2 < 2$ ; $63 \max_j t_j^2 < 1$          | $\max t_j = 3/25$ ;<br>$(1 + t^2)^2 = 401956/390625$ ;<br>$63t^2 = 567/625$ |
| (4) heights $ c_j /10000$ below $\rho = 1/100$<br>and $\eta = 1/100$ | $\max = 3/2500$                                                             |

## APPENDIX F. NOTATION

TABLE 9. Notation used throughout, with the section that defines it.

| symbol                            | meaning                                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $P$                               | the seven-cube chair carrier, volume 7 (Section 2)                                                                                   |
| panel                             | one of the 24 exposed unit squares of $P$ (Section 2)                                                                                |
| feature                           | one of the 192 square pyramids, eight per panel (Section 2)                                                                          |
| $\eta, \rho$                      | base half-width 1/100 of a feature; tube half-height and retained-core margin 1/100 (Sections 2 and E.5)                             |
| $Q$                               | the solid: the carrier with its 192 features (Section 2)                                                                             |
| role, profile                     | a feature position indexed by panel and mark; the vector of 192 signed feature coefficients (Section 2)                              |
| $t_j, \alpha_j, \beta_j$          | slope $j/100$ of a feature of magnitude $j$ ; its base-edge and ridge deviation angles (Lemma 4.3)                                   |
| $E(F)$                            | the closed edge graph of a feature $F$ : four base edges and four ridges (Section 4)                                                 |
| frame                             | a signed permutation of the coordinate axes; registered frames are the 24 proper ones (Section 3.1)                                  |
| registered pose                   | a proper cubic frame followed by an integer translation (Section 3.1)                                                                |
| $\mathcal{A}_{44}$                | the 44 legal neighbour poses relative to a root chair in the identity pose (Section 3.1)                                             |
| $F_0, F_1, \text{Partners}(u)$    | the 6,862 complete-feature companion poses; those not overlapping the root; those in $F_1$ matching the root feature $u$ (Section 5) |
| $T, \text{Per}(T), \text{Sym}(T)$ | a tiling; its translation periods; its symmetry group (Section 1.5)                                                                  |
| handedness                        | the determinant $\pm 1$ of the linear part of a placement (Section 7)                                                                |
| supertile of level $n$            | a $2^n$ -scaled chair in a registered pose (Section 6)                                                                               |
| coarsening $D$                    | grouping into unique parents, halving parent origins and scale, and realizing fresh copies of $Q$ (Theorem 6.5)                      |
| T1                                | kernel-checked Lean proof using only standard axioms                                                                                 |
| T1N                               | kernel-checked Lean proof with the recorded native compiler hooks                                                                    |
| T2                                | exact finite Python replay                                                                                                           |
| T3                                | written proof or cited external import                                                                                               |

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